

Biomarkers in Acute and Chronic Kidney Diseases

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CHAPTER OUTLINE

BIOMARKER DEFINITION, 873
PROCESS OF BIOMARKER DISCOVERY,
ASSAY VALIDATION, AND QUALIFICATION
IN A CLINICAL CONTEXT, 873
ANALYSIS OF BIOMARKER
PERFORMANCE, 874

CHARACTERISTICS OF AN IDEAL BIOMARKER
FOR KIDNEY DISEASE, 875
CRITICAL PATH INITIATIVE: A NEED FOR
BETTER BIOMARKERS, 902
KIDNEY HEALTH INITIATIVE, 903
FUTURE OF BIOMARKERS, 903

KEY POINTS

- An ideal biomarker is easily measurable, reproducible, sensitive, organ-specific, cost-effective, easily interpretable, and present in readily available specimens (e.g., blood and urine).
- Due to the inherent variability in the serum creatinine assay, patients with advanced chronic kidney disease (CKD) may be misclassified as acute kidney injury (AKI) based on small changes in creatinine levels.
- Serum cystatin C performs on par with serum creatinine for estimates of glomerular filtration rate (GFR) in the setting of CKD and may provide additional information about a CKD patient's risk for cardiovascular morbidity and mortality.
- Urinary α_1 -microglobulin, a low-molecular-weight glycoprotein and member of the lipocalin superfamily, has been associated with the increased risk of CKD and all-cause mortality in a variety of clinical settings.
- Heart-type fatty acid binding protein, proenkephalin, and monocyte chemoattractant protein-1 are promising AKI biomarkers in the setting of cardiac surgery and intensive care unit-associated AKI.
- Urinary TIMP-2 $\times$ IGFBP-7 is associated with the increased risk of development of Kidney Disease: Improving Global Outcomes stage 2 or 3 AKI in the next 12 hours in ICU patients at high risk for AKI.
- Plasma levels of *TGFR1*, *TGFR2*, EGF, and KIM-1 have been increasingly recognized as potential biomarkers for incident and progressive CKD.

Kidney disease is a global health problem. Acute kidney injury (AKI) and chronic kidney disease (CKD) are increasing in incidence.¹ In the United States, it is clear that the incidence of AKI, regardless of its severity, has been steadily increasing at a rate that is disturbingly high, and it is increasingly recognized that AKI predisposes to the progression of CKD toward end-stage renal disease (ESRD), which ultimately requires dialysis or kidney transplantation.^{2–4} According to the World Health Organization, approximately 850,000 patients develop ESRD every year.^{5–7} Across the globe, treatment of ESRD poses a major challenge for health care systems and the global economy. The burden of kidney disease is most significant in developing countries and is adversely influenced by inadequate socioeconomic and health care

infrastructures.^{5,8,9} Importantly, kidney disease progression may be curtailed if the disease is diagnosed early. Hence, detection and management of kidney diseases, whether acute or chronic, in the early, reversible, and potentially treatable stages, is of paramount importance. Biomarkers that will help diagnose kidney injury, predict progression of kidney disease, and provide information regarding the effectiveness of therapeutic intervention will be important adjuncts to our standard management strategies.

Recently, many novel, high-throughput technologies in the fields of genomics, proteomics, and metabolomics have made it easier to interrogate hundreds or even thousands of potential biomarkers at once, without prior knowledge of the underlying biology or pathophysiology of the system

being studied.^{10–13} As a result, there is a renewed interest in discovering novel biomarkers for use in drug development and patient care. Despite notable achievements, however, only a few biomarkers—blood urea nitrogen (BUN) level, creatinine concentration, urinalysis results and proteinuria—are routinely used to diagnose and monitor kidney injury. These commonly used gold standard biomarkers of kidney function are not optimal to detect injury or dysfunction early enough to allow prompt therapeutic intervention. Although additional candidate biomarkers have been reported, none have been adequately validated to justify their use in making patient care decisions, but a few look quite promising.

BIOMARKER DEFINITION

In 2001, the U.S. Food and Drug Administration (FDA) standardized the definition of a biomarker as “a characteristic that is objectively measured and evaluated as an indicator of normal biologic processes, pathogenic processes, or pharmacologic responses to therapeutic intervention.”¹⁴ The National Institutes of Health further classified biomarkers based on their utility (Table 27.1).¹⁴ Biomarkers can potentially

serve a wide range of functions in drug development, clinical trials, and therapeutic management strategies. There are many different classes of biomarkers—prognostic, predictive, pharmacodynamic, and surrogate biomarkers. Of note, these categories are not mutually exclusive. Definitions of the different types of biomarkers can be found in Table 27.1. Examples of biomarkers are proteins, lipids, genomic or proteomic patterns, imaging determinations, electrical signals, and cells present in urine. Some biomarkers also serve as surrogate endpoints. A surrogate endpoint is a biomarker intended to substitute for a clinical endpoint. Furthermore, a surrogate endpoint biomarker is expected to predict clinical benefit (harm or lack of benefit) based on epidemiologic, therapeutic, pathophysiologic, or other scientific evidence.¹⁵ An ideal biomarker is easily measurable, reproducible, sensitive, cost-effective, easily interpretable, and present in readily available specimens (blood and urine).

PROCESS OF BIOMARKER DISCOVERY, ASSAY VALIDATION, AND QUALIFICATION IN A CLINICAL CONTEXT

Primary challenges to the development of biomarkers for kidney injury and toxicity are discovery of candidate markers, design of an assay, validation of the assay, and qualification of the biomarker for use in specific clinical contexts. The process of biomarker identification and development is arduous and involves several phases.^{16,17} For the purpose of simplicity, this process can be divided into the following five phases (adapted and modified from Pepe and colleagues¹⁶).

PHASE 1: DISCOVERY OF POTENTIAL BIOMARKERS THROUGH UNBIASED OR HYPOTHESIS-GENERATING EXPLORATORY STUDIES

The primary goal of phase 1 is to identify potential leads using various technologies and to confirm and prioritize the identified leads. The search for biomarkers often begins with preclinical studies that compare either tissue or biologic fluids in diseased animals (e.g., animals with kidney injury) with those in healthy animals to identify genes or proteins that appear to be upregulated or downregulated in diseased tissue relative to control tissue. When biologic samples, such as blood and urine, are readily available from humans, it is possible to forgo the animal model stage. Innovative discovery technologies include microarray-based gene expression profiling that provides information regarding expression of genes, micro RNA (miRNA)-based expression, and proteomic technologies, as well as metabolomic, profiling of biologic fluids based on mass spectrophotometry, and other technologies. The candidate marker approach, especially when informed by the pathophysiology of the disease for which the biomarker is being evaluated, should not be ignored.

Once a promising biomarker is discovered, the validation process begins. An assay has to be developed and validated. The validation process is laborious and expensive, requiring access to patient samples with complete clinical annotation and long-term follow-up, as described in the section on phase 2. In addition, each biomarker must be qualified for

Table 27.1 Biomarker Definitions

Term	Definition
Biomarker	A characteristic that is objectively measured and evaluated as an indicator of a normal biologic process, pathogenic process, or pharmacologic response to therapeutic intervention. • A prognostic biomarker is a baseline patient or disease characteristic that categorizes patients by degree of risk for disease occurrence or progression, informing about the natural history of the disorder in the absence of a therapeutic intervention. • A predictive biomarker is a baseline characteristic that characterizes patients by their likelihood for response to a particular treatment, predicting either a favorable or unfavorable response. • A pharmacodynamic biomarker is a dynamic assessment that shows that a biologic response has occurred in a patient who has received a therapeutic intervention. Pharmacodynamic biomarkers may be treatment-specific or broadly informative of disease response, with the specific clinical setting determining how the biomarker is used and interpreted.
Clinical endpoint	A characteristic or variable that reflects how a patient fares or functions or how long a patient survives
Surrogate endpoint biomarker (type 2 biomarker)	A marker that is intended to substitute for the clinical endpoint. A surrogate endpoint is expected to predict clinical benefit, harm, lack of benefit, or lack of harm on the basis of epidemiologic, therapeutic, pathophysiologic, or other scientific evidence.

specific application. This is especially true in the case of kidney diseases, for which one biomarker alone may not satisfy the requirements of an ideal biomarker. This is described in the subsequent section on phase 4. Incorporation of several of these novel biomarkers into a biomarker panel may enable simultaneous assessment of site-specific kidney injury or several mechanisms contributing to clinical syndromes.

PHASE 2: DEVELOPMENT AND VALIDATION OF AN ASSAY FOR THE MEASUREMENT OR IDENTIFICATION OF THE BIOMARKER IN CLINICAL SAMPLES

The primary goal of phase 2 is to develop and validate a clinically useful assay that has the ability to distinguish a person with kidney disease or injury from a person with healthy kidneys in a high-throughput fashion. This phase involves development of an assay, optimization of assay performance, and evaluation of the reproducibility of the assay results within and among laboratories. Defining reference ranges of biomarker values is a crucial step before the biomarker can be used clinically.^{18,19} It is important to characterize how the levels of these markers vary with patient age, gender, race, and ethnicity, and how biomarker values are related to known risk factors.²⁰

PHASE 3: DEMONSTRATION OF THE BIOMARKER'S POTENTIAL CLINICAL UTILITY IN RETROSPECTIVE STUDIES

In phase 3, the primary objectives are to as follows: (1) evaluate the biomarker potential in samples obtained from a completed clinical study; (2) test the diagnostic potential of the biomarker for early detection; and (3) determine the sensitivity and specificity of the biomarker using defined threshold values of the biomarker for utility in prospective studies. For example, if the levels of biomarker differ significantly between cases (those with acute or chronic kidney injury) and control subjects only at the time of clinical diagnosis, then the biomarker shows little promise for population screening or early detection. In contrast, if levels differ significantly at hours, days, or years before clinical symptoms appear, then the biomarker's potential for early detection is increased. This phase also involves comparing the biomarker with several other novel biomarkers or existing gold standard biomarkers and defining the biomarker's performance characteristics (i.e., sensitivity, specificity) using receiver-operating characteristic curve analysis. This latter process is particularly challenging in kidney disease, given uncertainties in the sensitivity and specificity of the gold standard used.²¹

PHASE 4: PERFORMANCE OF PROSPECTIVE SCREENING STUDIES

The primary aim of phase 4 studies is to determine the operating characteristics of the biomarker in a relevant population by measuring detection and false referral rates. In contrast to phase 1, 2, and 3 studies, which are based primarily on stored specimens, studies in phase 4 involve screening subjects prospectively and demonstrating that clinical care is changed as a result of the information provided by the biomarker analysis.

BIOMARKER QUALIFICATION PROCESS

The application for FDA qualification of novel biomarkers requires the intended use of the biomarker in nonclinical and clinical contexts and the collection of evidence supporting qualification. This can be a joint and collaborative effort among regulatory agencies, pharmaceutical companies, and academic scientists.²²

Data are shared between the FDA and pharmaceutical industry or academic laboratories through voluntary exploratory data submissions (VXDSs).²³ Submission of exploratory biomarker data through VXDSs allows interaction between reviewers at the FDA and researchers in industry or academia regarding study designs, sample collection and storage protocols, technology platforms, and data analysis. This pilot process for biomarker qualification allows the Predictive Safety Testing Consortium to apply to both U.S. and European drug authorities simultaneously for qualification of new nephrotoxic biomarkers (e.g., kidney injury molecule-1, albumin, total protein, cystatin C, clusterin, trefoil factor 3, and α_2 -microglobulin) as predictors of drug-mediated nephrotoxicity.^{23–25} The FDA and the corresponding European authority (European Medicines Agency [EMA]) reviewed the application separately and made decisions as to whether each would allow the new biomarkers to be “fit for purpose” in preclinical research.^{24,25} Some of these markers were proposed to be qualified as biomarkers for clinical drug-induced nephrotoxicity once further supportive human data are submitted. More recently, the FDA has approved total kidney volume as a biomarker of disease progression for the purposes of clinical trial enrichment in autosomal dominant polycystic kidney disease (see also Chapter 45), thus opening to door to other novel measures as potential biomarkers of disease progression in the setting of acute kidney injury (AKI) and other forms of chronic kidney disease (CKD).^{22,26}

It is notable that the process described above is specific for the FDA and United States and that the biomarker validation and approval process varies significantly around the world. In the past few years, the FDA, EMA, and other agencies have approved several biomarkers for clinical use in the United States and other countries throughout Europe and Asia.

PHASE 5: CONTINUED ASSESSMENT OF THE VALIDITY OF THE BIOMARKER IN ROUTINE CLINICAL PRACTICE

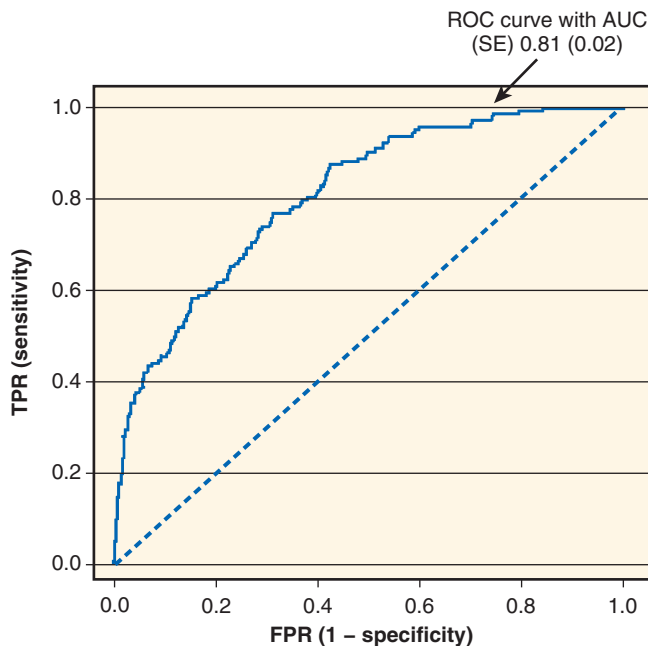
Phase 5 addresses whether measurement of the biomarker alters physician decision making and/or reduces the mortality or morbidity associated with the given disease in the population.

ANALYSIS OF BIOMARKER PERFORMANCE

The widely accepted measure of biomarker sensitivity and specificity is the receiver-operating characteristic (ROC) curve.²⁷ ROC curves display the proportion of subjects, with and without disease, correctly identified at various cutoff points. A ROC curve is a graphic display of trade-offs between the true-positive rate (sensitivity) and false-positive rate (1 – specificity, where specificity is expressed as a value

from 0 to 1) when the biomarker is a continuous variable (Fig. 27.1).^{28,29} Sensitivity is plotted along the ordinate, and the value of $(1 - \text{specificity})$ is plotted on the abscissa. Each point on the curve represents the true-positive rate and false-positive rate associated with a particular test value. The diagonal, represented by the equation true-positive rate (sensitivity) = false-positive rate $(1 - \text{specificity})$, corresponds to the set of points for which there is no selectivity in predicting disease. The area under this line of “unity” is 0.5, which indicates no advantage relative to the flip of a coin.

The performance of a biomarker can be quantified by calculating the area under the ROC curve (AUC). The AUC is the probability that a randomly sampled case has a larger biomarker value (or risk score) than a randomly sampled control. Although this makes the AUC easily interpretable, this interpretation is not always clinically meaningful, because cases and controls do not present to clinicians in random pairs. Thus, whereas an ideal biomarker could supply an AUC of 1.0 (a clinical rarity), in actuality the AUC lacks true direct clinical relevance.²⁹ Despite these flaws, the AUC is widely reported and familiar to clinicians. The shortcomings of AUC extend into the assessment of the incremental change in AUC (ΔAUC) when adding a new marker to a group of previously established predictors. The clinical impact of a ΔAUC of 0.02 is often unclear, and the statistics and *P* values behind such calculations remain problematic.^{30,31}



		True disease state	
		Diseased	Nondiseased
Biomarker test	Positive (diseased)	True-positive (TP)	False-positive (FP)
	Negative (nondisease)	False-negative (FN)	True-negative (TN)

Biomarker classification by disease status

True-positive rate (TPR) = sensitivity = $\text{TP}/(\text{TP} + \text{FN})$

False-positive rate (FPR) = $1 - \text{specificity} = \text{FP}/(\text{FP} + \text{TN})$

Fig. 27.1 Receiver operator characteristic curves.

Other important parameters related to biomarker performance, primarily with respect to the testing of larger or specific populations, are positive and negative predictive values. The positive predictive value is the proportion of persons who test positive for a disease and actually have the disease, whereas the negative predictive value represents the proportion of persons who test negative and do not have the disease. There is considerable interest in developing algorithms that use a composite of values of several biomarkers that are measured in parallel for the purpose of increasing diagnostic potential or predicting disease course and patient outcomes.

More recently, the Net Reclassification Index (NRI) and the Integrated Discrimination Improvement Indices (IDIs) have been used to evaluate the ability of new biomarkers. The NRI is simply the proportion of the population whose risk category changes with the new biomarker; a small reclassification rate means that treatment decisions will rarely be altered by the new biomarker. IDI is defined as the difference in discrimination slopes between the unadjusted and biomarker-adjusted clinic models, with large effect sizes having an IDI ≥ 0.10 and medium effect sizes having an IDI between 0.05 and 0.10.³² Note that the NRI and IDI have not been widely accepted by all statisticians.³³

CHARACTERISTICS OF AN IDEAL BIOMARKER FOR KIDNEY DISEASE

Characteristics of an ideal biomarker for kidney disease are described in Table 27.2. For AKI, the biomarker should have the following characteristics: (1) be organ-specific and allow differentiation between intrarenal, prerenal, and postrenal causes of AKI, as well as acute glomerular injury; (2) be able to detect AKI early in the course and predict its course and potentially the future implications of AKI; (3) be able to identify the cause of AKI; (4) be site-specific and able to inform pathologic changes in various segments of renal tubules during AKI, as well as correlate with the histologic findings in kidney biopsy specimens; (5) be easily and reliably measured in a noninvasive or minimally invasive manner; (6) be stable in its matrix; (7) be rapidly and reliably measurable at the bedside; and (8) be inexpensive to measure.

In CKD (unlike AKI), the timing and nature of the insult are very hard to estimate, which makes the search for early biomarkers for CKD very difficult. An ideal biomarker for CKD shares many of the same requirements described earlier for AKI biomarkers, including providing insight into the following: (1) the location of the injury (e.g., glomerular, interstitial, tubular); (2) the disease mechanism; (3) the progressive course of the disease; and (4) the risk of complications from comorbid conditions such as cardiovascular disease and diabetes.

ACUTE KIDNEY INJURY MARKERS

In the cardiac sciences, the discovery of biomarkers, such as troponins, which reflect early cardiomyocyte damage rather than decreased cardiac function, has enabled the development and implementation of novel therapeutic strategies to reduce coronary insufficiency and associated morbidity and mortality.^{34,35} By contrast, the delay in diagnosis associated

Table 27.2 Characteristics of an Ideal Kidney Biomarker**Functional Properties**

- Rapid and reliable increase in response to kidney diseases
- Highly sensitive and specific for acute and/or chronic kidney disease
- Shows good correlation with degree of renal injury
- Provides risk stratification and prognostic information (e.g., severity of kidney disease, need for dialysis, length of hospital stay, and mortality)
- Site-specific to detect early injury (e.g., proximal, distal, interstitium or vasculature) and identify pathologic changes in specific segments of renal tubules
- Applicable across different races and age groups
- Allows recognition of the cause of kidney injury or disease (e.g., ischemia, toxins, sepsis, cardiovascular disease, diabetic nephropathy, lupus, or combinations)
- Organ-specific; allows differentiation among intrarenal, prerenal, and extrarenal causes of kidney injury
- Noninvasively identifies the duration of kidney failure (acute kidney injury, chronic kidney injury)
- Useful to monitor the response to therapeutic interventions
- Provides information on the risk of complications from comorbid conditions (especially in chronic kidney disease)

Physiochemical Properties

- Stable over time across different temperature and pH conditions, with clinically relevant storage conditions
- Rapidly and easily measurable
- Not subject to interference by drugs and endogenous substances

Table 27.3 Kidney Disease: Improving Global Outcomes (KDIGO) Staging of Acute Kidney Disease

Stage	Serum Creatinine Criteria	Urine Output Criteria
1	1.5–1.9 times baseline or ≥0.3 mg/dL (26.5 μmol/L) increase	<0.5 mL/kg for 6–12 h
2	2.0–2.9 times baseline	<0.5 mL/kg/h for ≥12 h
3	≥3.0 times or Increase in serum creatinine to ≥4.0 mg/dL (≥353.6 μmol/L) or Initiation of renal replacement therapy or In patients <18 years, decrease in eGFR to <35 mL/min ²	<0.3 mL/kg/h for ≥24 h or anuria for 12 h

with the use of kidney biomarkers, such as serum creatinine concentration, has impaired the ability of nephrologists to conduct interventional studies in which the intervention can be implemented early in the course of the disease process.³⁶ Although the past decade has seen a revolution in terms of diagnostic criteria for AKI with the RIFLE (risk, injury, failure, loss, end-stage kidney disease) classification³⁷ and the Acute Kidney Injury Network (AKIN) definition of AKI³⁸ being harmonized into the Kidney Disease: Improving Global Outcomes (KDIGO) classification³⁹ (Table 27.3), these criteria remain limited by their reliance on the serum creatinine concentration on some level. More recently, there has been a call to expand these definitions further to potentially include biomarkers, but as of the time of this publication, these

new guidelines have yet to be widely accepted.⁴⁰ These new guidelines and the concept of AKI remain reliant on the serum creatinine level and will continue to serve as a limitation, given creatinine's role as a functional biomarker. The serum creatinine level can increase in cases of prerenal azotemia when there is no tubular injury and can be unchanged under conditions of significant tubular injury, particularly when patients have good underlying kidney function and significant kidney reserve. Nonetheless, these criteria have advanced our understanding of the epidemiology of AKI, and these standardized consensus definitions have allowed for comparisons and aggregation of data from a larger number of papers.⁴¹ Biomarkers of AKI can serve several purposes and are no longer thought of as a replacement for the serum creatinine level. Table 27.4 summarizes several of the potential uses of AKI biomarkers. Fig. 27.2 summarizes the kidney-specific location of the AKI biomarkers discussed later.

Urine and serum biomarkers each have advantages and disadvantages. Serum biomarkers are often not stable and are difficult to measure because of interference with several serum proteins. By contrast, urinary biomarkers are relatively stable and easy to assess; however, their concentrations are greatly influenced by the hydration and volume status of the patient and other conditions that affect urinary volume. To overcome this challenge, urinary biomarker concentrations have often been normalized to urinary creatinine concentrations to eliminate the influence of urinary volume on the assumption that the urinary creatinine excretion rate is constant over time, and that biomarker production or excretion has a linear relationship with the urinary creatinine excretion rate. Bonventre and colleagues have challenged this assumption, especially in AKI settings, when the urine creatinine excretion rate is not constant and changes over time, greatly influencing the normalized value of a putative urinary biomarker after normalization. They have suggested that the most accurate method to quantify biomarkers is the timed collection of urine samples to estimate the renal excretion rate⁴²; however, this approach is not practical for

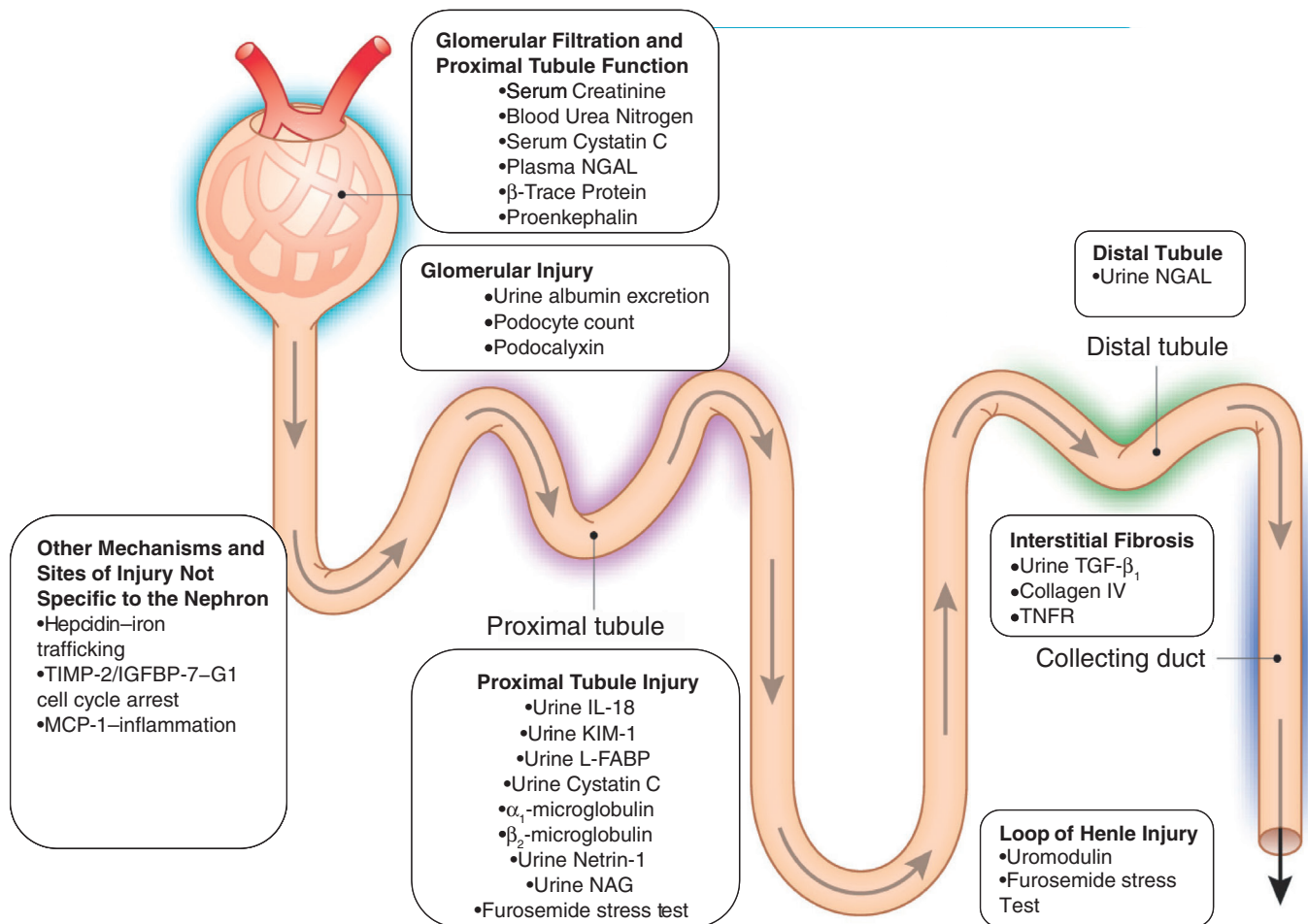


Fig. 27.2 Biomarkers in relation to their site of injury in the nephron. *GST*, Glutathione S-transferase; *IGFBP7*, insulin-like growth factor binding protein-7; *IL-18*, interleukin-18; *KIM-1*, kidney injury molecule-1; *L-FABP*, liver-type fatty acid binding protein; *MCP-1*, monocyte chemotactic protein 1; *NAG*, N-acetyl-β-D-glucosaminidase; *NGAL*, neutrophil gelatinase-associated lipocalin; *TGF-β₁*, transforming growth factor-β₁; *TIMP-2*, tissue inhibitor metalloproteinase-2; *TNFR*, tumor necrosis factor receptor. (Adapted from Koyner JL, Parikh CR. Clinical utility of biomarkers of AKI in cardiac surgery and critical illness. *Clin J Am Soc Nephrol*. 2013;8:1034–1042.)

Table 27.4 Potential Uses for Biomarkers of Acute Kidney Injury (AKI) and Chronic Kidney Disease (CKD)

Disorder	Potential Use
AKI	Early detection of AKI: • Differential diagnosis of AKI (e.g., distinguishing between volume-mediated AKI [prerenal] and intrinsic tubular injury (e.g., acute tubular necrosis)) • Predicting outcomes of AKI at the time of clinical diagnosis (need for RRT, development of post-AKI CKD, short- and long-term mortality) • Predicting recovery from AKI • Ascertaining the nephron-specific location and cause of renal injury
CKD	Monitoring the effects of an intervention Early detection and diagnosis of CKD: • Predicting the progression of CKD (rapid vs. slow progression) • Predicting outcomes of CKD at the time of clinical diagnosis (development of ESRD, short- and long-term mortality) • Predicting cardiovascular disease and outcomes in patients with CKD • Monitoring the effects of an intervention

routine clinical care. Endre and colleagues delved into this issue further by demonstrating that the ideal method for quantitating biomarkers of urinary AKI depends on the outcome of interest; absolute biomarker concentrations best diagnosed AKI at the time of intensive care unit (ICU) admission, whereas normalization to urinary creatinine improved the prediction of incipient AKI.⁴³ A potential explanation of the failings of normalization is that it will often amplify the signal. For example, when the glomerular filtration rate (GFR) is reduced in immediate response to a tubular injury, the amount of biomarker produced will increase, and the urinary creatinine level will decrease. The normalized value will therefore increase by a greater amount in the short term than can be explained by the increase in the absolute level of biomarker production. Currently, there is no standardized method of accounting for this issue, with some urinary biomarkers being normalized to urine creatinine and others being reported without normalization.

Because AKI and CKD share functional and structural aspects, there are overlapping as well as distinct classes of functional and structural biomarkers. Among the functional markers, the GFR is often used as the gold standard. Although the true GFR, as determined by agents that are freely filtered and undergo minimal handling by the tubule (e.g.,

Table 27.5 Biomarker Performance in Detecting Acute Kidney Injury (AKI)^a

Parameter	Perioperative AKI				Critically Ill			Emergency Room	
	Preop AKI Risk	Early Postop AKI	AKI Progression	Long-Term Mortality	Early Diagnosis of AKI	Type of AKI (Transient vs. Intrinsic)	Need for RRT	Early Diagnosis of AKI	Type of AKI (Transient vs. Intrinsic)
Urine NGAL	N/A	+	—	+	+	+	+	+	+
Blood NGAL	—	+	+	+	—	?	—	?	?
Blood CysC	+	+	—	?	+	+	+	?	?
Urine CysC	N/A	—	—	—	+	+	+	+	+
Urine IL-18	N/A	+	+	+	+	+	+	+	+
Urine KIM-1	N/A	+	—	+	+	—	—	+	+
Urine L-FABP	N/A	—	—	+	?	?	—	+	+
TIMP-2 IGFBP-7	N/A	+	+	?	+	?	+	+	?
Urine protein, albumin	+	+	+	+	?	?	?	?	?

^aFrom multicenter studies at a variety of clinical time points.

CysC, Cystatin C; IL-18, interleukin-18; KIM-1, kidney injury molecule-1; L-FABP, liver fatty acid binding protein; NGAL, neutrophil gelatinase-associated lipocalin; Preop, preoperative; Postop, postoperative; RRT, renal replacement therapy; TIMP-2, IGFBP-7, tissue inhibitor metalloprotease 2, insulin-like growth factor binding protein-7.

+, Data published display the ability to detect this aspect of AKI; —, data published do not display the ability to detect this aspect of AKI; ?, no large multicenter data published on this biomarker or aspect of AKI; N/A, Not applicable to the following: (1) biomarkers of tubular injury have no role in preoperative risk screening; (2) serum creatinine is intrinsic to the definitions of AKI being tested.

Adapted and expanded from Koyner JL, Parikh CR. Clinical utility of biomarkers of AKI in cardiac surgery and critical illness. *Clin J Am Soc Nephrol*. 2013;8:1034–1042.

iothalamate, iohexol, inulin), represents a sensitive measure for determining changes in kidney function, these tests are invasive and laborious to perform. Moreover, because of the renal reserve, changes in GFR may not indicate structural injury until significant injury has occurred. On the other hand, structural markers of tubular injury are expressed by tubular cells, and subtle changes in epithelial cells lead to release of these markers into the urine. It is becoming increasingly clear that many of these biomarkers serve as signals for both AKI and CKD and also may be used to monitor progression from AKI to CKD. A challenge is to define at what level of release of these markers the injury is clinically significant in either the acute or chronic setting. Failure to identify the separate impacts of CKD and AKI on these biomarker values will lead to inappropriate clinical decision and/or poor results in clinical studies.^{44,45}

Table 27.5 summarizes the ability of biomarkers to detect clinical endpoints related to AKI in a variety of clinical settings.

GLOMERULAR INJURY MARKERS

SERUM GLOMERULAR FILTRATION MARKERS

During the course of injury, kidney function may be impaired with reduction in the GFR and accumulation of several nitrogenous compounds in the blood. Serum creatinine and BUN concentrations are routinely used as markers of kidney injury, but it is important to recognize these parameters as markers for kidney dysfunction, rather than direct markers of injury.

As discussed elsewhere in this text, the estimated GFR (eGFR), using creatinine as a biomarker, is most reliable for CKD under steady-state conditions. In the acute setting, its use is more problematic for reasons that have already been discussed. In healthy persons, the GFR is in the range of 90

to 130 mL/min/1.73 m². By definition, patients with stage 4 or 5 CKD have GFRs that are below 30 mL/min/1.73 m².⁴⁶ Complications of CKD are more pronounced at lower GFRs, and mild to moderate CKD may progress to end-stage renal disease (ESRD).

In AKI, the GFR is only indirectly linked to kidney injury, and changes in the GFR reflect a late consequence in a sequence of events associated with a primary insult to the kidney. Furthermore, because of renal reserve, a large amount of functioning renal tissue can be lost without significant changes in the GFR.^{47,48} The functional effects of renal reserve on the GFR can be demonstrated in kidney donors, who often have only modest changes in serum creatinine levels and GFR after donating one kidney, even though 50% of the renal mass is lost.⁴⁹

Ideally, a serum GFR marker should be freely filtered, with no reabsorption or secretion in the tubule, and should maintain a constant plasma level when kidney function is stable. GFR can be determined using exogenous and endogenous markers of filtration. Evaluation of the GFR using the exogenous markers inulin, iothalamate, or iohexol provides reliable results and represents the gold standard; however, the process is time-consuming and expensive and can be performed only in specialized settings.⁴⁶ Once the GFR level falls below 60 mL/min/1.73 m², renal functional impairment can be estimated adequately by the serum creatinine level using various equations to calculate the eGFR.^{50–52} Although traditionally these equations have been less accurate for those with higher GFRs, newer formulas have been constructed using more patients with normal and near-normal GFRs.⁵²

Creatinine

Determination of the eGFR using endogenous creatinine is cost-effective but can be problematic. Creatinine is a

breakdown product of creatine and phosphocreatine, which are involved in the energy metabolism of skeletal muscle. Creatinine is freely filtered by the glomerulus, but is also to a lesser degree (10%–30%) secreted by the proximal tubule. Under normal conditions, the daily synthesis of creatinine of approximately 20 mg/kg of body weight reflects muscle mass and varies little.⁵³

Accumulated data from various studies have indicated that the creatinine concentration is not an ideal marker for diagnosing AKI for a variety of reasons, including the following^{54–56}:

1. Creatinine production and its release into the circulation vary greatly with age, gender, muscle mass, certain disease states and, to a lesser extent, diet. For example, in rhabdomyolysis, serum creatinine concentrations may rise more rapidly due to the release of preformed creatinine from the damaged muscle. Also, body creatinine production, as measured by 24-hour urinary excretion, decreases with older age, falling from a mean of 23.8 mg/kg of body weight in men aged 20 to 29 years to 9.8 mg/kg of body weight in men aged 90 to 99 years, largely because of the reduction in muscle mass.⁵⁷
2. Serum creatinine concentrations are not specific for renal tubular injury. For example, intravascular volume depletion “prerenal” factors (e.g., severe dehydration, blood volume loss, altered vasomotor tone, age-related decrease in renal blood flow) and postrenal factors (e.g., obstruction or extravasation of urine into the peritoneal cavity) may falsely elevate serum concentrations in the absence of parenchymal damage. Thus, a decrease in the eGFR inferred from an increase in the serum creatinine level may not distinguish among prerenal, intrinsic renal, and postrenal causes of impaired kidney function, which may not be the case for some biomarkers of renal tubular injury.⁵⁸ Even in cases in which the serum creatinine level is elevated as a consequence of direct renal injury, it cannot be used to determine the location of the injury (glomerular vs. tubular, or proximal tubular vs. distal tubular).
3. Static measurement of the serum creatinine level does not reflect the real-time changes in the GFR resulting from acute changes in kidney function as creatinine accumulates over time. Given the large amounts of functional kidney reserve in healthy persons and the variable amounts of kidney reserve in patients with mild to moderate disease, the creatinine level is not a sensitive marker.⁵⁹
4. Drug-induced reduction in tubular secretion of creatinine might result in underestimation of kidney function. Medications such as cimetidine and trimethoprim inhibit creatinine secretion and increase the serum creatinine concentration without affecting the true GFR.^{60,61}
5. One commonly used creatinine assay—the Jaffe method—is subject to interference by intake of certain drugs or by certain pathophysiologic states, including hyperbilirubinemia and diabetic ketoacidosis.⁶⁰
6. Small changes in serum creatinine levels are subject to high false-positive rates, and this effect is magnified in those with a higher baseline creatinine due to the inherent variability in the creatinine assay. Thus patients with CKD may be misclassified as having AKI based solely on small changes in the serum creatinine level.⁶²

Similarly, the use of serum creatinine levels in CKD is limited by several patient-dependent and independent variables, including age, race, sex, and comorbid conditions. Serum creatinine concentration can significantly decrease in advanced kidney disease, unrelated to its renal clearance.⁶³ The sensitivity of serum creatinine levels in determining kidney function can be improved by serial measurements of timed creatinine clearance (usually, but not always, 24-hour collections). However, many individuals find this collection cumbersome, and errors (e.g., skipped voids) typically lead to underestimation of function.

The serum level creatinine is stable during long-term storage, after repeated thawing and refreezing,⁶⁴ and for up to 24 hours in clotted whole blood at room temperature.⁶⁵ The Jaffe reaction-based assay (alkaline picrate assay) is routinely used in clinical laboratories to assess creatinine levels. However, Jaffe methods overestimate serum creatinine concentration by approximately 25% due to the interference of noncreatinine chromogens, particularly proteins. Interference from glucose^{66,67} and acetoacetate⁶⁸ are particularly important because diabetic patients are particularly prone to develop CKD. As a result, eGFRs are higher when Jaffe methods are used than when other approaches are used. Expert professional bodies have recommended that all methods of creatinine measurement should become traceable to a reference method based on isotope dilution mass spectrometry.⁶⁹ Several modifications of the Jaffe method have been made to increase the specificity by decreasing the influence of interfering substances.^{70,71} Enzymatic methods of measuring creatinine have been widely adopted by clinical laboratories as an alternative to alkaline picrate assays. Although various substances do interfere with enzymatic assays, the assays are reported to be subject to less interference than Jaffe methods.^{72–74} The high-performance liquid chromatography (HPLC)-based assay has evolved as a potential alternative approach for measurement of the serum creatinine level.^{75,76} Several studies have demonstrated that HPLC methods have greater analytic specificity than conventional methods.^{77–79} This approach clearly has severe limitations with respect to throughput, however.

Finally, over the past decade, there has been a dedicated national effort (within the United States) to standardize serum creatinine assays by establishing calibration traceability to an isotope dilution mass spectrometry (IDMS) reference standard. Prior to standardization, there was a large variability in serum creatinine results among clinical laboratories, with roughly a 10% to 20% bias being reported in the literature.⁸⁰ This process, which started in 2005 and has recently been completed, has led to the standardization of assays, which has led to less variation and more accurate eGFR measurements when used in conjunction with the IDMS-traceable estimating equations.^{81,82}

Blood Urea Nitrogen

Blood urea is a low-molecular-weight waste product derived from dietary protein catabolism and tissue protein turnover, and its levels are inversely correlated with decline in the GFR.⁸³ Urea is filtered freely, and a variable amount (≈30%–70%) is reabsorbed predominantly in the proximal tubule, with recycling between the tubule and interstitium in the kidney medulla. The normal range of urea nitrogen in blood or serum is 5 to 20 mg/dL (1.8–7.2 mmol urea/L).⁸³

The wide reference range reflects the influence on BUN of nonrenal factors, including dietary protein intake, endogenous protein catabolism, fluid intake, and hepatic urea synthesis.^{83,84} BUN concentrations also increase with excessive tissue catabolism, especially in cases of fever, severe burns, trauma, high corticosteroid dosage, chronic liver disease, and sepsis.⁸³ In addition, any factor that increases the tubular reabsorption of urea, including decreased effective arterial volume (i.e., impaired renal perfusion) and/or obstruction of urinary drainage, will increase the BUN concentration.^{83,85,86} Because of these limitations, BUN is not a sensitive and specific marker for acute or chronic kidney disease. However, for those with advanced CKD (e.g., CKD 4–5), some have suggested averaging urea clearance and creatinine clearance to serve as a more accurate estimate of the true GFR. This is in part because at these lower levels of renal function, creatinine clearance will overestimate the (secretion) GFR, whereas urea clearance will underestimate the GFR.⁸⁷ BUN is measured by spectrophotometry. Because of these undesirable limitations of creatinine and BUN as markers, there has been a great deal of interest in the identification of improved biomarkers for kidney injury.

Cystatin C

For the past 10 to 15 years, there has been a tremendous amount of research investigating serum cystatin C as a marker of GFR, and urinary cystatin C excretion has been proposed as a tubular injury marker. In 1961, Butler and Flynn studied the urine proteins of 223 individuals by starch gel electrophoresis and found a new urine protein fraction in the post- γ -globulin fraction.⁸⁸ They named this protein fraction “cystatin C.” Cystatin C is a low-molecular-weight protein produced at a constant rate by all nucleated cells and eliminated exclusively by glomerular filtration. It has a small size (13 kDa) and a positive charge at physiologic pH. It is neither secreted nor reabsorbed by renal tubules but undergoes almost complete catabolism by proximal tubular cells; thus little, if any, appears in the urine under normal circumstances. Any impairment of reabsorption in proximal tubules can lead to marked increases in urinary levels of cystatin C in humans and animals. There have been a number of studies on the diagnostic potential of both serum and urinary cystatin C levels in acute and chronic kidney disease in humans.

Chronic Kidney Disease. Because of its short half-life (≈ 2 hours) and other properties described earlier, some believe that serum cystatin C levels reflect the GFR better than creatinine concentration. Initially, it was thought that the serum levels of cystatin C would be unaffected by gender, age, race, and muscle mass but, over the past several years, multiple studies have demonstrated that these factors are associated with altered levels of the biomarker.^{89,90} Notably, cystatin C levels have been shown to be associated with factors similar to those associated with creatinine—namely, that these levels may be elevated in males, taller and heavier patients, and those with higher lean body mass.^{89–91} However, unlike the serum creatinine level, which is usually lower in older adults, given their decreased muscle mass, a study investigating a subset of over 7500 subjects from the National Health and Nutritional Examination Survey (NHANES) III has demonstrated that cystatin C levels are elevated in more than 50% of those older than 80 years.⁹¹

Despite these minor limitations, cystatin C remains an excellent biomarker of CKD and performs on par, if not better than, the serum creatinine level in some cases. Equations for estimating GFR and CKD classification are discussed elsewhere in this text (Chapter 23). In a prospective cohort study of 26,643 Americans enrolled in the Reasons for Geographic and Racial Differences in Stroke (REGARDS) study, Peralta and colleagues have demonstrated that the cystatin C–based eGFR improves CKD classification and definition, as well risk stratification (for the development of ESRD and death) relative to a creatinine-based eGFR.⁹² This correlation with mortality was not novel because cystatin C demonstrated a stronger risk relationship with mortality than creatinine concentration or eGFR in older adults with cardiovascular disease.⁹³ In the Cardiovascular Health Study cohort of 4637 community-dwelling older adults, higher serum cystatin C concentrations were associated with a significantly elevated risk of death from cardiovascular causes (hazard ratio [HR], 2.27 [1.73–2.97]), myocardial infarction (HR, 1.48 [1.08–2.02]), and stroke (HR, 1.47 [1.09–1.96]) after multivariate adjustment. In the same study, higher serum creatinine values were not independently associated with any of these three outcomes.⁹⁴ Furthermore, a study in the general population has suggested that the cystatin C level has a stronger association with cardiovascular disease outcomes than creatinine concentration or eGFR, especially among older adults.^{94–96} On the other hand, a study based in primary care has reported that the use of cystatin C to estimate GFR in a predominantly older population with relatively mild CKD resulted in reclassification of a small proportion (7.7%) of participants with CKD category G3aA1, defined by creatinine to GFR (>60 mL/min/1.73 m²), but a much greater proportion (59%) were reclassified to a more advanced CKD category without any improvement in risk prediction.⁹⁷ Thus, serum cystatin C levels may be a better marker of kidney function than serum creatinine concentration in some, but not all, persons.

In addition to older adults, cystatin C has proven superior to serum creatinine in those infected with HIV. Choi and colleagues have demonstrated that the cystatin C–based eGFR outperformed serum creatinine in the ability to predict 5-year all-cause mortality in a cohort of 922 HIV-infected individuals.⁹⁸ This study mirrors recent findings from a study of 908 HIV-infected women, which demonstrated that CKD risk factors are associated with an overestimation of GFR by the serum creatinine level relative to cystatin C and that cystatin C significantly improves mortality risk prediction when added to a clinical model that already includes the serum creatinine level.⁹⁹

The concept of using cystatin C in concert, rather than in place of, serum creatinine has been gaining momentum.¹⁰⁰ Using a cross-sectional analyses and data from 5352 participants from 13 previously published studies, Inker and colleagues developed estimation equations using cystatin C alone and cystatin C and creatinine combined.¹⁰⁰ They then went on to validate these equations in a cohort of 1119 participants from 5 different studies. They demonstrated that combined equations outperformed the creatinine-alone equations and, in some cases, led to a NRI of 0.194 ($P < .001$). Although this study was performed predominantly in white subjects, thus limiting its broad applicability, it demonstrates the potential of cystatin C (and other biomarkers) to be used to augment the diagnostic scope of the serum creatinine

Box 27.1 Cystatin C

1. Serum cystatin C performs on par, if not, better than serum creatinine for the identification of patients with chronic kidney disease.
2. Serum cystatin C has been specifically shown to outperform serum creatinine in some older adults as well as those with HIV.
3. Additionally, serum cystatin C outperforms serum creatinine for its ability to detect those at risk for cardiovascular morbidity and mortality.
4. Serum cystatin C should be incorporated into the management and care of patients with chronic kidney disease.

level rather than replace it. Subsequently, the combined creatinine and cystatin C Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation has been shown to outperform each individual equation (creatinine and cystatin C) in a cohort of 805 older community-based individuals.¹⁰¹ Given the mounting clinical evidence and the emergence of automated assays and cheaper reagent costs (e.g., \$4), it is increasingly apparent that cystatin C should become a routine part of the nephrologist laboratory assessment of CKD and points to an increased role for cystatin C in the management of patients with CKD. However, as with any biomarker, its limitations should be considered when interpreting results (Box 27.1).¹⁰²

Acute Kidney Injury. Given its success as a marker of glomerular filtration, several groups have investigated serum cystatin C as a potential biomarker of AKI. In a single-center mixed ICU population of 85 subjects (44 of whom developed RIFLE Classified AKI), Herget-Rosenthal and colleagues demonstrated that serum cystatin C had excellent diagnostic value, predicting AKI 24 and 48 hours prior to serum creatinine (AUC of 0.97 and 0.82, respectively).¹⁰³ These data were followed up by a study of 442 patients from 2 separate ICUs that demonstrated that plasma cystatin C increased earlier than serum creatinine and was able to significantly predict several adverse patient outcomes, including sustained AKI, death, and dialysis.¹⁰⁴ Similarly, in a study of 202 diverse intensive care unit (ICU) patients, of whom 49 developed AKI based on urine output and/or serum creatinine RIFLE F criteria, serum cystatin C levels showed excellent predictive value for AKI. However, the serum cystatin C concentration did not rise earlier than the serum creatinine concentration.¹⁰⁵

Outside the ICU, cystatin C levels were shown to be capable of detecting a decrease in the GFR after contrast agent administration earlier than the serum creatinine value in adult patients who underwent coronary angiography.¹⁰⁶ In a prospective study of 87 patients who underwent elective catheterization, contrast medium-induced nephropathy occurred in 18 patients, and ROC analysis showed a higher AUC for cystatin C level than for serum creatinine concentration (0.933 vs. 0.832; $P = .012$). When a cutoff value of more than 1.2 mg/L was used, cystatin C level before catheterization exhibited 94.7% sensitivity and 84.8% specificity for predicting contrast medium-induced nephropathy.¹⁰⁷

Serum cystatin C has been studied as a biomarker for both early AKI (rising earlier than serum creatinine) and

AKI severity, with several smaller studies providing mixed results.^{108–111} More recently, the larger multicenter Translational Research Investigating Biomarker Endpoints in AKI (TRIBE-AKI) study investigated several aspects of serum cystatin C following both adult and pediatric cardiac surgery. In 1147 adults, Shlipak and colleagues demonstrated that preoperative serum cystatin C values outperformed serum creatinine- and creatinine-based eGFRs in its ability to forecast postoperative AKI.¹¹² After adjustment for clinical variables known to contribute to AKI, serum cystatin C had a C-statistic of 0.70 and an NRI of 0.21 compared with serum creatinine ($P < .001$). However, when this same group investigated sensitivity and rapidity of AKI detection (defined as a 25%, 50%, and 100% increase from preoperative values) by postoperative changes in serum cystatin C, they did not demonstrate a clear advantage over changes in the serum creatinine level.¹¹³ In a follow-up analysis, they demonstrated that postoperative elevations of cystatin C ($>25\%$) were associated with an increased risk of death during a 3-year follow-up period. This long-term mortality risk was higher in those with changes in cystatin C alone (adjusted HR, 2.2 [1.09–4.47] compared with those with changes in just the serum creatinine level (HR, 1.50 [0.96–2.34])).¹¹⁴ This mixed message of postoperative serum cystatin C in adults is in stark contrast to the results in 288 children undergoing cardiac surgery. Zappitelli and colleagues have demonstrated that serum cystatin C measured within the first 6 postoperative hours is associated with both stages 1 and 2 pediatric AKI.¹¹⁵ Additionally, postoperative serum cystatin C values were also associated with adverse patient outcomes, including length of mechanical ventilation and length of ICU stay. However unlike the adult population, preoperative values were not associated postoperative AKI.

 β -Trace Protein

β -Trace protein (BTP), also referred as “prostaglandin D synthase,” has emerged as another promising biomarker for the GFR. BTP is a small protein with a molecular weight of 23 to 29 kDa, depending on the size of the glycosyl moiety. BTP belongs to the lipocalin protein family, whose members are primarily involved in the binding and transport of small hydrophobic ligands. It is primarily produced in the cerebral fluid, where its concentrations are more than 40-fold higher than in the serum. BTP is primarily eliminated by glomerular filtration, and its concentrations in urine range from 600 to 1200 $\mu\text{g/L}$.¹¹⁶

The first observation of elevated BTP levels in association with impaired kidney function was reported by Hoffmann and associates in 1997.¹¹⁷ Since then, several research studies have been conducted to evaluate the sensitivity and specificity of BTP as a marker of GFR and to compare it with the serum creatinine level in patients with CKD¹¹⁸ and kidney transplant recipients. In two separate cohort studies, one adult and one pediatric, serum cystatin C was shown to outperform BTP for the detection of decreased renal function (as measured by inulin clearance); both markers were shown to outperform the serum creatinine level alone.^{119,120}

Foster and colleagues have investigated the association of BTP, serum cystatin C, and creatinine-based eGFR with all-cause mortality in a subset of patients from the NHANES cohort.¹²¹ They analyzed data from 6445 adults (enrolled from 1988–1994), with follow-up through December 2006. All three markers were associated with increased mortality

after adjusting for demographics. However, when comparing the mortality risk of the fifth (highest) quintile with the third (middle) quintile, only BTP (HR, 95% confidence interval [CI], 2.14 [1.56–2.94] and serum cystatin C (HR, 1.94 [1.43–2.62]) remained statistically significant (creatinine eGFR, HR 1.31 [0.84–2.04]). These effects remained significant when looking at either cardiovascular disease- or coronary heart disease-associated mortality. Similarly, in data from the Atherosclerosis Risk in Communities (ARIC) study, BTP was shown to outperform creatinine-based eGFR (CKD-EPI) in the prediction of mortality and development of kidney failure.¹²² More recently, BTP was investigated in a pooled cross-sectional analysis of patients from three separate studies, the Modification of Diet in Renal Disease (MDRD), African American Study of Kidney Disease (AASK), and Chronic Renal Insufficiency Cohort (CRIC).¹²³ In 3156 patients with CKD, BTP levels were strongly associated with the serum creatinine level and urine protein excretion and demonstrated a correlation with race, where levels were lower in African Americans. Additionally, BTP was clearly associated with advanced age and male gender and inversely related to body mass index.¹²³

BTP has also been investigated for its ability to predict and estimate residual kidney function in those undergoing maintenance hemodialysis.^{124,125} Using interdialytic urine collections and serum BTP and β_2 -microglobulin levels, Wong and colleagues derived equations to estimate GFR and residual urea clearance in 191 patients on hemodialysis.¹²⁵ Using the reciprocal of BTP and other factors, they demonstrated an R^2 of 0.70 with GFR and an R^2 of 0.625 with residual urea clearance. Similarly, Shafi and colleagues developed separate equations in 44 dialysis patients completing 24-hour urine tests and went on to validate their equations in a cohort of 826 patients from the Netherlands Cooperative Study on the Adequacy of Dialysis.¹²⁴ They demonstrated that BTP provided better accuracy than the traditional markers—creatinine and urea clearance—and that BTP had an AUC of 0.82 for detecting the gold standard—measured urinary urea clearance. Thus, BTP and other markers show promise as a method to estimate residual renal function in dialysis patients.

Concentrations of BTP are not affected by commonly used immunosuppressive medications, such as prednisone, mycophenolate mofetil, and cyclosporine.¹²⁶ This is especially useful when evaluating kidney function in kidney transplant recipients, in whom cystatin C concentrations may be falsely elevated due to steroid treatment.¹²⁷ Unlike for serum creatinine values, age and race were not associated with BTP concentrations. Several GFR estimation equations based on BTP have been developed for use in kidney transplant recipients.^{126,127} However, these equations, similar to those discussed previously for dialysis patients, will require external validation in larger and more diverse patient groups. In contrast to creatinine, one limitation of using BTP is lack of widespread availability and standardization of the assay.

URINARY GLOMERULAR CELL INJURY MARKERS

Defects in podocyte structure have been reported in many glomerular diseases, which have been classified as “podocytopathies.”^{128,129} Injured podocytes have been reported in immunologic and nonimmunologic forms of human glomerular disease, including hemodynamic injury, protein

overload states, injury from environmental toxins, minimal change disease, focal segmental glomerulosclerosis, membranous glomerulopathy, diabetic nephropathy, and lupus nephritis.^{130–135} Podocytes may be injured in many forms of human and experimental primary glomerular disease and in secondary forms of focal segmental glomerulosclerosis, including those caused by hypertension, diabetes, and tubulointerstitial disease.^{136–138} Before detachment from the glomerular basement membrane, podocytes undergo structural changes, including effacement of foot processes and microvillous transformation.^{128,129,139,140}

PODOCYTE COUNT

After undergoing the aforementioned structural changes, podocytes detach from the glomerular basement membrane and are excreted into the urine. Urinary levels of viable podocytes have been extensively studied in several renal diseases.^{141–144} Numerous studies have reported that the number of podocytes shed in patients with active glomerular disease is significantly higher than in healthy controls and in patients with inactive disease. Importantly, the podocyte number in urine correlates with disease activity (assessed by renal biopsy) and has been shown to decline with treatment.¹⁴⁵ Studies have linked podocytopenia and disease severity in immunoglobulin (IgA) nephropathy^{141,142} and diabetic nephropathy.^{143,144} An improved and standardized laboratory method is urgently needed to facilitate measurement of urinary podocyte number. Alternative methods that indirectly assess the number of podocytes in urine include detection of messenger RNA (mRNA) and protein levels of podocyte-specific proteins by polymerase chain reaction (PCR) assay and enzyme-linked immunosorbent assay (ELISA), respectively.

PODOCALYXIN

Podocalyxin is the most commonly used marker protein for detecting podocytes in urine.¹⁴⁶ It is a highly *O*-glycosylated and sialylated type I transmembrane protein of approximately 140 kDa and is expressed in podocytes, hematopoietic progenitor cells, vascular endothelial cells, and a subset of neurons.¹⁴⁶ Podocalyxin participates in a number of cellular functions through its association with the actin cytoskeleton, ezrin, and Na^+/H^+ exchanger regulatory factor 1 and 2 (NHERF1 and NHERF2) proteins. Urinary podocalyxin has been reported as a marker of activity in a number of diseases, including IgA nephropathy, Henoch-Schönlein purpura, diabetic nephropathy, lupus nephritis, poststreptococcal glomerulonephritis, membranous nephropathy, focal segmental glomerulosclerosis, and preeclampsia.^{147–155} Unfortunately, because podocalyxin is expressed on a number of cell types, the presence of podocalyxin in the urine is not always reflective of urinary podocytes.

URINARY TUBULAR INJURY MARKERS

Microscopic examination of the urine has been used for many years to gain insight into the degree of glomerular and tubular injury. Other components of the urine have been used to quantitate tubular cell injury in a more specific and sensitive fashion. These markers have been demonstrated to be extremely valuable in detecting kidney injury in the setting of AKI. Moreover, some of these biomarkers, such as

interleukin-18 (IL-18), kidney injury molecule-1 (KIM-1), neutrophil gelatinase-associated lipocalin (NGAL), and liver-type fatty acid binding protein (L-FABP), have been shown to be potentially useful in a variety of contexts in acute and chronic kidney injury. In this section, the utility of urine microscopy is described briefly, and some of the emerging biomarkers of tubular injury are discussed.

URINE MICROSCOPY

Urine microscopy with sediment examination is a time-honored test that is routinely used to assist in the diagnosis of kidney injury.^{156–158} The urine from patients with tubular injury typically contains proximal tubular epithelial cells, proximal tubule epithelial cells casts, granular casts, and mixed cellular casts. Patients with predominantly prerenal azotemia occasionally have hyaline or fine granular casts in their urine.^{159–161} Several studies have shown that the increase in urinary cast excretion correlates well with AKI.^{160,162,163} Marcussen and associates¹⁶² have demonstrated that patients with tubular injury have a high number of granular casts compared with those with prerenal azotemia.

There has been a resurgence of urinalysis sediment scoring systems for the diagnosis of AKI.^{161,164} Several of these systems have shown excellent specificity for AKI and correlate well with the severity of AKI.^{164–166} However, their widespread acceptance has been hampered by the relatively modest sensitivity of urine microscopy for detecting AKI.^{161,164,167,168} Urine microscopy remains a user-dependent tool that displays a tremendous amount of interphysician variability, which likely contributes to its suboptimal sensitivity for AKI.¹⁶⁹ Three of the most widely reported urine microscopy scoring systems are reviewed in Table 27.6.

Several other studies have looked at the potential of using urine microscopy in combination with other biomarkers for detecting tubular injury, with varying degrees of success.^{165–167} In the near future, urine microscopy, a current mainstay in

the clinical diagnosis of AKI, could be used in concert with markers of glomerular function and validated biomarkers of tubular injury to diagnose AKI.

α_1 -MICROGLOBULIN

α_1 -Microglobulin is a low-molecular-weight glycoprotein of approximately 27 to 30 kDa and is a member of the lipocalin superfamily. α_1 -Microglobulin is primarily synthesized by the liver and is available in free form and as a complex with IgA.¹⁷⁰ α_1 -Microglobulin has been detected in human serum, urine, and cerebrospinal fluid. Urine and serum levels have been found to be elevated in patients with renal tubular diseases. α_1 -Microglobulin is freely filtered at the glomerulus and completely reabsorbed and catabolized by the normal proximal tubule. Megalin mediates the uptake of this protein in the proximal tubule. Therefore, an increase in the urinary concentration of α_1 -microglobulin indicates proximal tubular injury or dysfunction.

The urinary levels of α_1 -microglobulin are influenced by age. The normal range in populations younger than 50 years of age is less than 13 mg/g of creatinine and, in those 50 years of age and older, is less than 20 mg/g of creatinine.¹⁷⁰ Unlike β_2 -microglobulin, α_1 -microglobulin is more stable over a range of pH levels in the urine,¹⁷¹ which makes it a more acceptable urinary biomarker.

Acute Kidney Injury

α_1 -Microglobulin quantitation in the urine has been reported as a sensitive biomarker for proximal tubule dysfunction in adults and children.^{170,172} In a small cohort of 73 patients, of whom 26 required renal replacement therapy, Herget-Rosenthal and colleagues compared levels of α_1 -microglobulin, β_2 -microglobulin, cystatin C, retinol binding protein, α -glutathione S-transferase, lactate dehydrogenase, and *N*-acetyl- β -D-glucosaminidase (NAG) early in the course of AKI.¹⁷³ They found that urinary cystatin C and α_1 -microglobulin had the highest ability to predict the need for renal replacement therapy. In this study, urinary α_1 -microglobulin had an AUC of 0.86 for predicting the need for renal replacement therapy. This is similar to the results of a Zheng and colleagues,¹⁷⁴ who measured α_1 -microglobulin levels in 58 children undergoing cardiac surgery and found that levels were higher in those who developed AKI (AKIN criteria). Four hours after cardiopulmonary bypass, α_1 -microglobulin had an AUC of 0.84 (0.72–0.95), with a value of 290 mg/g providing a sensitivity of 90% and specificity of 79%. However, follow-up studies have reported mixed results, with Martensson and colleagues¹⁷⁵ finding no difference in α_1 -microglobulin levels between those with and without AKI in the setting of sepsis and septic shock in a small prospective single-center study of 45 subjects.

More recently, α_1 -microglobulin levels at the time of arrival to the emergency room (ER) have demonstrated the ability to correlate with the development of AKI, with an AUC of 0.88 and a cutoff value of 35 mg/g providing reasonable sensitivity (80%) and specificity (81%). However, α_1 -microglobulin did not remain an independent predictor of AKI in the multivariable model (OR [95% CI], 1.85 [0.80–4.31]).¹⁷⁶ In addition, α_1 -microglobulin has also been reported as a useful marker for proximal tubular damage, and recovery in early infancy and has been shown to correlate with tubular atrophy and interstitial fibrosis on renal transplant biopsy 1 year after transplantation.^{177,178}

Table 27.6 Review of Urine Microscopy Scoring Systems

Study	Scoring System
Chawla et al. 2008 ¹⁶⁴	Grade 1: No casts or RTE Grade 2: At least one cast or RTE but <10% of LPF Grade 3: Many casts or RTEs (from 10%–90% of LPF) Grade 4: Sheet of muddy brown casts and RTEs in >90% of LPF
Perazella et al. 2010 ¹⁶⁵	0 points: No casts or RTE seen 1 point each: One to five casts/LPF or one to five RTEs/HPF 2 points each: Six or more casts/LPF or $\geq$ six RTEs/HPF
Bagshaw et al. 2011 ¹⁶⁷	0 points: No casts or RTE seen 1 point each: One cast or one RTE/HPF 2 points each: Two to four casts or RTEs/HPF 3 points each: Five casts or more or five or more RTEs/HPF

HPF, High-power field; LPF, low-power field; RTE, renal tubule epithelial cells.

Chronic Kidney Disease

Increasingly, α_1 -microglobulin has been investigated in the setting of CKD. Limited studies have demonstrated that this marker may correlate with disease activity and proximal tubule damage in the setting of diabetic nephropathy, as well as idiopathic membranous nephropathy.^{179,180} In a cohort of 2948 Framingham Heart Study participants, O'Seaghdha and colleagues¹⁸¹ demonstrated that although α_1 -microglobulin does not correlate with the development of CKD or albuminuria during the mean of 10.1 years follow-up, it does correlate with all-cause mortality (HR, 1.26 [1.13–1.40; $P < .001$]). In a different cohort, Jotwani and colleagues have demonstrated that α_1 -microglobulin levels in women infected with human immunodeficiency virus (HIV) are associated with both kidney function decline and mortality.¹⁸² Compared with those with the lowest α_1 -microglobulin levels, those with the highest levels had a 2.1 (1.3–3.4)-fold increased risk of developing CKD and a 2.7-fold risk of a 10% decline of their eGFR. This correlation in CKD development and progression was separate from the 1.6-fold adjusted risk of mortality, which accounted for baseline kidney function as well as the presence of albuminuria.¹⁸² Follow-up studies in this same cohort did not link α_1 -microglobulin levels with an *APOL1* genotype.¹⁸³

Despite these recent data on urine concentration, the use of α_1 -microglobulin remains limited due to prior studies demonstrating variation in serum levels with age, gender,¹⁸⁴ and clinical conditions, including liver disease,¹⁷⁰ ulcerative colitis,¹⁸⁵ HIV infection, and mood disorders,¹⁷⁰ as well as the lack of international standardization. Urinary α_1 -microglobulin is measured by an immunonephelometric assay.

β_2 -MICROGLOBULIN

β_2 -Microglobulin is a low-molecular-weight polypeptide with a molecular weight of 11.8 kDa. β_2 -Microglobulin is present on the cell surface of all nucleated cells and in most biologic fluids, including serum, urine, and synovial fluid. β_2 -Microglobulin is normally excreted by glomerular filtration, reabsorbed almost completely ($\approx 99\%$), and catabolized by the normal proximal tubule in humans.^{186,187} Megalin mediates the uptake of this protein in the proximal tubule.¹⁸⁷ In healthy individuals, approximately 150 to 200 mg of β_2 -microglobulin is synthesized daily, with a normal serum concentration of 1.5 to 3 mg/L. Any pathologic state that affects kidney tubule function will result in an increase in β_2 -microglobulin levels in the urine because of the impeded uptake of β_2 -microglobulin by renal tubular cells. For spot urine collections, the concentration of β_2 -microglobulin in healthy individuals is typically 160 $\mu\text{g/L}$ or less or 300 $\mu\text{g/g}$ of creatinine or less. Unlike urea, its serum levels are not influenced by food intake, which makes it an attractive marker for malnourished patients with low serum urea levels. In patients with CKD, increases in serum β_2 -microglobulin levels reflect a decrease in glomerular function. In the aforementioned cross-sectional study of patients from the MDRD, AASK, and CRIC studies, serum β_2 -microglobulin levels were weakly associated when male gender and non-African American race were compared; however, unlike other markers, it was strongly associated with smoking. Thus, serum β_2 -microglobulin is clearly affected by nonrenal variables, demonstrating some of the limitations when implementing β_2 -microglobulin as a marker of GFR.¹²³

In ESRD patients, serum levels of β_2 -microglobulin are usually in the range of 20 to 50 mg/L. β_2 -Microglobulin

accumulation is linked to toxicity because the molecule precipitates and forms fibrillary structures and amyloid deposits, particularly in bone and periarticular tissue, which leads to the development of carpal tunnel syndrome and erosive arthritis.^{187,188} Elevated serum levels of β_2 -microglobulin have been reported in several AKI and CKD clinical settings, including cadmium toxicity,¹⁸⁹ following cardiac surgery,^{190,191} liver transplantation,¹⁹² and renal transplantation.¹⁹³ In idiopathic membranous nephropathy, the β_2 -microglobulin level was identified as a superior independent predictor of the development of GFR decline.¹⁹⁴ Other studies have reported that β_2 -microglobulin performs as well as, if not better than, the serum creatinine level for the detection of AKI in critically ill children¹⁹⁵ or following adult cardiac surgery.¹⁹¹

Serum concentrations of β_2 -microglobulin should be interpreted cautiously because they are altered significantly in various diseases, including rheumatoid disorders and several types of cancers.^{196,197} Initially, it was believed that the increase in β_2 -microglobulin levels in CKD is solely due to declines in kidney function, but some studies have shown that other factors, including increased synthesis of β_2 -microglobulin, may contribute in patients with ESRD.¹⁹⁸ Another significant drawback associated with the use of urinary β_2 -microglobulin as a marker of kidney injury is its instability in the urine at room temperature, particularly when the pH is less than 5.5. Because of this, the urine should be alkalized and frozen at -80°C immediately after collection.^{188,199}

HEPCIDIN-25

Hepcidin-25 is a 2.8-kDa hormonal regulator of iron metabolism produced in the liver, heart, and kidney. Hepcidin-25 binds to and induces the internalization and degradation of the transmembrane iron exporter ferroportin.²⁰⁰ Hepcidin-25 acts to downregulate iron uptake and reduce extracellular iron availability from stored iron.²⁰¹ Given its link to iron metabolism, and the fact that free iron is known to be released in the setting of the ischemia reperfusion injury and oxidative stress that occur with cardiopulmonary bypass, urinary hepcidin-25 has been investigated as a marker of kidney injury following cardiac surgery. Ho and colleagues²⁰² have identified urinary hepcidin-25 in a nested case-control study of 44 adults who underwent cardiac surgery. Using surface-enhanced laser desorption/ionization time-of-flight mass spectrometry (SELDI-TOF-MS), urine from 22 individuals who developed at least RIFLE risk AKI and 22 individuals whose creatinine level did not increase more than 10% from baseline during the postoperative period (no AKI), they demonstrated that hepcidin-25 is dramatically upregulated in the urine of those with no AKI. Taking this a step further, this same group quantified the concentration of hepcidin-25 in the urine (normalized to urine creatinine) and demonstrated that concentrations were higher in those who did not go on to develop postoperative AKI ($P < .0005$).

In a multivariable analysis, hepcidin-25 was significantly associated with the avoidance of AKI, with urinary concentrations on postoperative day 1 showing an AUC of 0.80.²⁰³ The data from this small study have been corroborated by another modestly sized cohort of 100 adults undergoing cardiopulmonary bypass (CPB). Haase-Fielitz and colleagues have demonstrated that 6 hours after CPB, urinary hepcidin-25 levels are lower in the 9 subjects who developed RIFLE-based

AKI compared with those with no AKI (AUC = 0.80; $P = .004$).²⁰⁴ More recently, Prowle and colleagues measured hepcidin-25 in a 93-subject multicenter cohort of patients undergoing cardiac surgery with CPB.²⁰⁵ Although this study did not have a 6-hour CPB time point, it did demonstrate that hepcidin-25 values 24 hours after surgery were different in those with and without AKI ($P < .001$), with values being higher in those without AKI. Additionally, they demonstrated that combining hepcidin-25 with postoperative NGAL values leads to an AUC of 0.84 for the development of RIFLE-based AKI. In addition to establishing a universal commercially available assay, these results warrant further preliminary investigations in other AKI settings, as well as prompt validation in a larger cohort of cardiac surgery patients.

INTERLEUKIN-18

IL-18 is an 18-kDa proinflammatory cytokine that is activated by caspase-1 and produced by renal tubule cells and macrophages. Animal studies have indicated that IL-18 is a mediator of acute tubular injury, including both neutrophil and monocyte infiltration of the renal parenchyma.^{206,207} In studies using caspase-1 knockout mice, these mice experienced the same degree of ischemic AKI as wild-type mice injected with an IL-18 neutralizing antiserum, demonstrating that IL-18 is an important mediator of ischemic AKI.²⁰⁷

Other studies have shown that IL-18 plays a major role in macrophage activation, with mice engrafted with IL-18-deficient bone marrow experiencing a lower incidence of AKI than those with IL-18-replete marrow.²⁰⁸ Similarly, in IL-18 knockout mice with AKI, tumor necrosis factor- α (TNF- α), inducible nitric oxide synthase, macrophage inflammatory protein-2, and monocyte chemoattractant protein-1 messenger RNA expression are all decreased, indicating the deleterious impact of IL-18 on AKI. In the human kidney, IL-18 is induced and cleaved mainly in the proximal tubules and released into the urine. IL-18 has been shown to participate in a variety of renal disease processes, including ischemia reperfusion injury, allograft rejection, infection, autoimmune conditions, and malignancy. IL-18 is easily and reliably measured in the urine by commercially available ELISA and microbead-based assays.

Acute Kidney Injury

Several studies have demonstrated the usefulness of IL-18 as a biomarker for the detection of AKI. Originally, Parikh and associates studied a group of 72 patients and reported urinary IL-18 levels to be significantly higher in patients diagnosed with acute tubular necrosis than in patients with prerenal azotemia or urinary infection or in healthy control subjects with normal renal function.²⁰⁹ Since then, several large multicenter studies have gone on to investigate the ability of IL-18 to detect AKI in a variety of clinical settings.

The Translational Research Investigating Biomarker Endpoints in AKI (TRIBE-AKI) consortium measured IL-18 in 1219 adults who underwent cardiac surgery. The study selected those at high risk for postoperative AKI because of the presence of one of the following: emergency surgery, preoperative serum creatinine level >2 mg/dL, left ventricular ejection fraction $<35\%$, stage III or IV New York Heart Association cardiac failure, age >70 years, preexisting diabetes mellitus, concomitant coronary artery bypass grafting (CABG), and valve surgery or repeat cardiac surgery. Those

with preoperative AKI, kidney transplants, ESRD, or a preoperative serum creatinine level >4.5 mg/dL were excluded. After dividing the cohort into quintiles, the highest quintile of IL-18 was associated with a 6.8-fold higher risk of AKI, defined as a postoperative doubling of the serum creatinine level or undergoing acute dialysis.²¹⁰ The first postoperative concentration of IL-18 (0–6 hours) had an AUC of 0.74 that increased to 0.76 after combining IL-18 with a clinical model of factors known to affect AKI risk. The TRIBE-AKI pediatric cohort (311 children) reported results in line with the TRIBE adult study, in which the highest quintile was associated with 9.4-fold increased risk of developing AKI (doubling or dialysis) compared with the first quintile. The effect was slightly attenuated after adjusting for clinical factors known to affect AKI (adjusted odds ratio [OR], 95% CI, 6.9 [1.7–28.8]).²¹¹

Urine IL-18 displayed a similar trend when investigating the association between biomarker concentrations and duration of AKI.²¹² Stratifying the cohort of those with AKI according to the duration, 61.7% ($n = 251$) had a duration of 1 to 2 days, 28.9% ($n = 118$) had a duration of 3 to 6 days, and 9.3% ($n = 38$) had a duration >7 days. The fifth quintile of IL-18 was strongly associated with the duration of AKI, providing an unadjusted OR of 3.90 (2.62–5.78). Although this effect was attenuated after adjusting for clinical factors known to affect AKI, it remained significant, at 2.90 (1.80–4.68).²¹²

In another secondary analysis of the TRIBE-AKI adult cohort, Koynert and colleagues²¹³ have demonstrated that IL-18 at the time of AKI can forecast those with early AKI (AKIN stage 1) who go on to progress to more severe stages of AKI (AKIN stage 2 or 3). In the 380 adults who developed at least stage 1 AKI, 45 went on to progress to stage 2 or 3. In the entire cohort, the fifth quintile of IL-18 was at increased risk for developing progressive AKI (3.63 [1.64–.03]), and this effect was only slightly attenuated after adjusting for the clinical model (3.00 [1.25–7.25]).

When investigated in a separate secondary analysis of this cohort, IL-18 concentrations collected in the immediate postoperative period were associated with long-term mortality following cardiac surgery. This investigation provided a median follow-up of 3.0 years (interquartile range [IQR], 2.2–3.6) during which 139 of the 1199 subjects died (50 deaths/1000 person-years). After adjusting for clinical factors known to affect mortality, patients without AKI ($n = 792$) in the third tertile of IL-18 were at increased risk of long-term mortality compared with the first tertile (adjusted HR, 1.23 [1.02–1.48]). This effect was magnified in subjects with perioperative AKI ($n = 407$), with the third tertile having an adjusted HR of 3.16 (1.53–6.53) compared with the reference cohort. Thus, IL-18 provides additional prognostic information about long-term postoperative mortality in patients with and without AKI.²¹⁴

In the setting of critical illness and ICU admission, IL-18 has not demonstrated the same robust results. In a study of 451 critically ill subjects, 86 of whom developed AKI within the first 48 hours, Siew and colleagues have demonstrated that urine IL-18 does not reliably predict AKI.²¹⁵ Although IL-18 levels at the time of ICU admission were higher in those who went on to develop AKI, the AUC was 0.62 (0.54–0.69) and only marginally improved to 0.67 after the exclusion of those with prior known CKD (eGFR < 75 mL/min). Despite the inability to detect AKI reliably, urine IL-18

levels did correlate with other adverse patient outcomes, including the need for renal replacement therapy (RRT) and 28-day mortality. This poor performance in the setting of critical illness has been corroborated by others, including data from a post hoc analysis of the EARLY ARF trial. In this prospective observational study in two large general ICUs ($n = 529$), IL-18 had an AUC of only 0.62 for the diagnosis of AKIN stage 1 AKI, but once again performed much better at forecasting the need for RRT or death (within 7 days). Unlike other biomarker studies, this IL-18 study did not demonstrate improved predictive powers, with the cohort stratified according to preadmission CKD stage.²¹⁶ In a separate post hoc analysis of this same EARLY ARF cohort, urinary IL-18 concentrations were shown to be significantly higher in those with prerenal azotemia ($n = 61$, defined as AKI that recovered within 48 hours of ICU admission, and was associated with a fractional excretion of sodium $<1\%$), compared with those with no AKI ($n = 285$). There was a trend toward higher values in those with AKI ($n = 114$, nonprerenal) compared with those with prerenal AKI ($P = .053$).²¹⁷

In a single-center study of 339 mixed surgical and medical ICU patients, Doi and colleagues have demonstrated that IL-18 levels are significantly elevated in those with both established and newly diagnosed AKI at the time of ICU arrival.²¹⁸ Although biomarker concentrations and AUCs were higher in those with established AKI (AUC, 0.78) compared with newly diagnosed AKI (AUC, 0.59), both these subgroups were significantly different compared with the no-AKI cohort. In this same study, IL-18 levels were significantly higher in nonsurvivors. These results were fairly similar to those of Nickolas and colleagues,²¹⁹ who measured urinary biomarkers in 1635 ER patients at the time of admission and compared the values to adjudicated AKI outcomes, in which prerenal AKI was defined as RIFLE Risk that returned to baseline within 72 hours and in the clinical setting, suggested decreased transient effective circulating volume. They demonstrated that IL-18 values for those with more severe intrinsic AKI were significantly higher compared with those with prerenal AKI. However, there was no difference between those with prerenal AKI and no AKI.

At kidney transplantation, the IL-18 level accurately identified delayed graft function (DGF) (AUC = 0.90) and predicted the rate of increase in serum creatinine concentration.²²⁰ To understand the utility of IL-18 and urinary NGAL in predicting graft recovery after kidney transplantation, Hall and colleagues²²² conducted a prospective, multicenter, observational cohort study of recipients of deceased donor kidney transplants. They collected serial urine samples from 91 patients for 3 days after transplantation. After adjustment for recipient and donor age, cold ischemia time, urine output, and serum creatinine concentration, NGAL and IL-18 concentrations accurately predicted the need for dialysis in transplant recipients. Furthermore, NGAL and IL-18 concentrations also predicted graft recovery up to 3 months later.²²² In further follow-up of this cohort, urine IL-18 concentrations collected at the time of surgery were correlated with 1-year posttransplantation graft outcomes. Above-median values of IL-18 on the first postoperative day had an adjusted OR of 5.5 (1.4–21.5, 95% CI) with poor graft function, defined as a GFR < 30 mL/min or return to RRT.²²³

In a study by Ling and associates involving patients who underwent coronary angiography, urinary IL-18 and NGAL

concentrations were significantly increased at 24 hours postprocedure in those who developed contrast-induced nephropathy, but not in the control group.²²⁴ ROC curve analysis demonstrated that both IL-18 and NGAL showed better performance in the early diagnosis of contrast nephropathy than the serum creatinine level ($P < .05$). Importantly, elevated urinary IL-18 concentrations 24 hours after contrast administration were also found to be an independent predictive marker for later major cardiac events (relative risk, 95% CI, 2.09 [1.15–3.77]).

Chronic Kidney Disease

There are promising data about IL-18 in the setting of CKD. In patients with diabetic kidney disease and proteinuria, IL-18 levels in renal tubular cells are higher than in patients with nondiabetic proteinuric disease.²²¹ In the Women's Interagency HIV study, urine IL-18 levels were independently associated with a more rapid loss of renal function after multivariable adjustment.²²⁵ In this cohort study of 908 HIV-infected women, urine IL-18 was the only biomarker (KIM-1 and albumin-to-creatinine ratio [ACR]) were also measured) that was associated with worsening renal function over time, as measured by eGFR–cystatin C. Urine IL-18 predicted an increased relative risk of renal function decline between 1.4 and 2.16, depending on the models used. In a follow-up study, the same group measured urine IL-18 in 908 HIV-infected and 289 noninfected women in the Women's Interagency HIV Study.²²⁶ In this cross-sectional cohort study, they demonstrated that after multivariable adjusted linear regression analysis that IL-18 concentrations are significantly higher in those with HIV (38%; $P < .0001$). They subsequently demonstrated this same phenomenon in a cross-sectional study of 813 HIV-infected men and 331 uninfected men, with urine IL-18 being 52% higher in those with HIV.²²⁷ Returning the female cohort, they demonstrated that urine IL-18 concentrations are significantly associated with higher HIV RNA levels and lower CD4 counts, hepatitis C infection, and high-density lipoprotein cholesterol levels, thus indicating a more extensive role for IL-18 in the setting of HIV-related kidney care.²²⁶ This promising HIV data are in contrast to those of the Consortium for Radiologic Imaging for the Study of Polycystic Kidney Disease (CRISP), which measured IL-18 in 107 patients with autosomal dominant polycystic kidney disease and found that although there was an increased mean IL-18 over the 3-year follow-up period, there was no association between tertiles of IL-18 and change in total kidney volume or eGFR.²²⁸

KIDNEY INJURY MOLECULE-1

Kidney injury molecule-1 (KIM-1 in humans, Kim-1 in rodents), which is also referred as “T-cell immunoglobulin and mucin domain-containing protein-1” (TIM-1) and “hepatitis A virus cellular receptor-1” (HAVCR-1), is a type I transmembrane glycoprotein with an ectodomain containing a six-cysteine immunoglobulin-like domains, two N-glycosylation sites, and a mucin domain. In an effort to identify molecules involved in kidney injury, Ichimura and colleagues originally discovered Kim-1 using representational difference analysis (a PCR-based technique) in rat models of acute ischemic kidney injury.^{229,230} Importantly, KIM-1 was shown to be significantly expressed in kidneys, specifically in proximal tubular cells of humans after ischemic injury, whereas it was virtually

absent or present at low levels in healthy kidneys. KIM-1 has evolved as a marker of proximal tubular injury, the hallmark of virtually all proteinuric, toxic, and ischemic renal diseases. KIM-1 has been shown to be a highly sensitive and specific marker of kidney injury in several rodent models, including models of injury due to ischemia,^{229,231} cisplatin, folic acid, gentamicin, mercury, chromium,^{232,233} cadmium,²³⁴ contrast agents,²³⁵ cyclosporine,²³⁶ ochratoxin A, aristolochic acid, D-serine, and protein overload.²³⁷

In 2002, Han and associates²³¹ published the first clinical study linking urinary levels of KIM-1 with AKI, demonstrating that tissue expression of KIM-1 is correlated with the severity of acute tubular necrosis and corresponding levels of KIM-1 ectodomain in the urine of patients with clinically significant AKI. Since then, numerous other studies have been published on the ability of KIM-1 to detect AKI in a variety of settings, including cardiac surgery, critical illness, and general hospitalized AKI, with mixed success.^{44,216,217,238–244}

KIM-1 has been investigated in several multicenter larger trials. In the TRIBE-AKI adult and pediatric cardiac surgery cohorts, the fifth quintile of urinary KIM-1 values was associated with an increased risk of AKI (those with postoperative doubling were at a 6.2-fold increased risk of AKI compared with those with the lowest KIM-1 values). This risk remained significant (4.8-fold) after adjusting for the clinical model—age, race, gender, CPB time, nonelective surgery, preoperative GFR, diabetes, hypertension, and study center. The effect was completely attenuated after including urinary IL-18 and plasma and urine NGAL into the model.²⁴³ Urine KIM-1 concentrations were associated with duration of AKI (categories: 1 to 2 days, 3 to 6 days, and >7 days). Compared with the first quintile, the fifth quintile of early postoperative KIM-1 provided an unadjusted OR of 2.96 (2.01–4.37) for longer duration of AKI. After adjusting for clinical factors known to affect AKI (including but not limited to age, gender, race, CPB time, baseline kidney function, diabetes, and hypertension), the OR remained statistically significant 2.30 (1.51–3.53).²¹²

When looking at long-term mortality in the TRIBE cohort, the third tertile of perioperative urine KIM-1 concentrations was associated with increased mortality. In those without AKI ($n = 792$), the third tertile had an adjusted HR of 1.83 (1.44–2.33) for mortality, whereas in the 407 subjects with AKI, the adjusted HR was slightly higher 2.01 (1.31–3.1).²¹⁴ In the pediatric cohort, those in the fifth quintile of urine KIM-1 were at increased risk of AKI in the unadjusted analysis; however, this effect was attenuated and no longer significant after multivariable adjustment.²⁴³

The data on urine KIM-1 in the setting of critical illness-related AKI have been just as mixed as the perioperative results, with KIM-1 having an AUC of 0.66 (95% CI; 0.61–0.72) for the diagnosis of AKI in samples from the EARLY ARF study. The results were less impressive with regard to the prediction of dialysis (0.62) or death (0.56) within the first week following ICU admission.²¹⁶ On the other hand, in a cohort of 529 mixed ICU patients, urine KIM-1 outperformed other biomarkers in its ability to detect AKI at the time of ICU admission in those with a preadmission GFR < 60 mL/min (AUC = 0.70, [0.58–0.82]). In a separate post hoc analysis of this cohort, KIM-1 levels demonstrated a significant stepwise increase when comparing those with no AKI (median, 170 µg/mmol creatinine [Cr]; interquartile range [IQR], 69–445) to

those with prerenal AKI (291 µg/mmol Cr; IQR, 121–549) and those with intrinsic AKI (lasting more than 48 hours; 376 µg/mmol Cr; IQR, 169–943).²¹⁷ In addition, urine KIM-1 was able to forecast the development of intrinsic AKI at the time of ER arrival with an AUC of 0.71 (0.65–0.76; $P < .001$). KIM-1 values again increased in a stepwise fashion, with values being lowest in those with no AKI or CKD < stable CKD < prerenal AKI < intrinsic AKI. Also, KIM-1 values were able to forecast the need for RRT, as well as inpatient mortality.²¹⁹

The usefulness of KIM-1 has been demonstrated not only as a urinary marker but also as a tool for evaluating kidney injury in kidney biopsy specimens by immunohistochemical methods. For example, Van Timmeren and associates found that the level of KIM-1 protein expression in proximal tubule cells correlated with tubulointerstitial fibrosis and inflammation in kidney tissue specimens from 102 patients who underwent kidney biopsy for a variety of kidney diseases.²⁴⁵ In a subset of patients whose urine was collected near the time of biopsy, urinary KIM-1 levels were significantly correlated with tissue KIM-1 expression in biopsy samples from patients with deterioration in kidney function and histologic changes indicative of tubular damage. In biopsy specimens from transplanted kidneys, increased KIM-1 staining was seen in 100% of patients with a deterioration of kidney function and pathologic changes indicating tubular injury, in 92% of patients with acute cellular rejection, and in 28% of patients with normal biopsy findings.²⁴⁶ This is in contrast to the findings of Hall and associates, who have demonstrated that urinary KIM-1 levels do not correlate with early posttransplantation or 1-year graft function.^{223,241} Similarly, Schroppel and colleagues have investigated KIM-1 RNA expression in pretransplantation biopsies collected from both living and deceased donor kidneys and found no significant correlation between KIM-1 staining and the occurrence of DGF.²⁴⁷

Chronic Kidney Disease

KIM-1 also shows promise as a useful biomarker in CKD. In addition to serving as a marker of proximal tubule dysfunction, animal studies have demonstrated that KIM-1 is upregulated in the later phases of AKI and plays an important role in renal repair; thus, it may be a major player in the pathophysiology of repair and CKD development after AKI.²⁴⁸ The ability to serve as a marker of CKD was evident in a nested case-control study from 686 participants from the Multi-Ethnic Study of Atherosclerosis (MESA). Cases were defined as those with a baseline eGFR > 60 mL/min who subsequently developed CKD stage 3 and/or had a rapid drop in kidney function over the 5-year study period. Each doubling of KIM-1 level (in pg/mL) was associated with a 1.15 (1.02–1.29) increased odds of developing CKD stage 3 or a rapid decline in GFR.

Similarly, at study entry, those in the highest decile of KIM-1 had a twofold increased risk of this same endpoint compared with the lower 90%. This ability to predict the development and progression of CKD was independent of the presence of albuminuria.²⁴⁹ Similarly, when investigated in a cohort of 149 persons with chronic congestive heart failure during 5 years of follow-up, KIM-1 levels were strongly associated with the progression of CKD (defined as a >25% drop in eGFR from baseline; $P < .05$).²⁵⁰ Urine KIM-1 and NAG were both independent predictors of the combined endpoint of CKD progression and all-cause mortality.²⁵⁰ In a multivariable Cox proportional hazard model that included

age, gender, body mass index (BMI), history of diabetes, hypertension, stroke, baseline eGFR, albuminuria, left ventricular ejection fraction, diuretic use, and brain natriuretic peptide levels, baseline eGFR, diuretic use, and KIM-1 and NAG levels were the only independent predictors of both these outcomes. More recently, in a pooled multivariable-adjusted analysis of five unique cohort studies, log-transformed urine KIM-1 levels (normalized to urine creatinine) were found to be higher in current smokers, and lower in blacks and those patients receiving angiotensin-converting enzyme inhibitors (ACE-Is) or angiotensin receptor blockers (ARBs).²⁵¹ Similarly, KIM-1 levels were higher in those with lower eGFRs and higher in those with albuminuria.

These results, which indicate the utility of KIM-1 as a biomarker for CKD, are in contrast to those of Bhavsar and colleagues, who measured KIM-1 in a case-control substudy of the ARIC study.²⁵² In this study of 286 subjects, 143 of whom developed new-onset CKD stage 3, KIM-1 did not display the ability to forecast or diagnose those at risk for CKD development or progression. In a follow-up case-control study by Foster and colleagues of 135 patients with ESRD and 186 controls from the same ARIC cohort (study visit 4), it was found that urinary KIM-1, normalized to urine creatinine, was associated with an increased risk of ESRD (unadjusted OR, 2.24 [1.97–4.69]; $P = .03$).²⁵³ However, this effect was attenuated when adjusting for other factors that increase the risk for ESRD (e.g., age, baseline albuminuria, baseline eGFR).

A recent study of the 2466 participants in the Chronic Renal Insufficiency Cohort Study (CRIC) reported findings similar to those of Foster and colleagues²⁵³ on the role of KIM-1 as a biomarker of CKD progression.²⁵⁴ In this study, with 9433 person-year follow-up, there were 581 cases of CKD progression defined as incident ESRD or a 50% decrease in eGFR. All four upper quintiles of KIM-1 were associated with increased risk of progression (compared with the lowest quintile) in unadjusted analyses (e.g., fifth quintile HR of 7.68 [5.61–10.5]). However, this effect was completely attenuated after the addition of other variables known to associate with CKD progression (e.g., age, gender, race, albuminuria, diabetes, cardiovascular disease).²⁵⁴

KIM-1 has been investigated and shown promise in a variety of other clinical settings, including in children with chronic renal tubular damage from vesicoureteral reflux,²⁵ HIV nephropathy,^{225,256} posttransplantation DGF,²⁵⁷ and diabetic nephropathy.^{258,259} In patients with IgA nephropathy, urinary KIM-1 levels were significantly higher than in healthy controls. Furthermore, the levels of urinary KIM-1 correlated positively with serum creatinine concentration and proteinuria and correlated inversely with creatinine clearance. Similarly, tubular KIM-1 expression, as determined by immunohistochemical analysis, correlated closely with urinary levels ($r = 0.553$; $P = .032$).²⁶⁰ Sundaram and associates have evaluated the potential of KIM-1, L-FABP, NAG, NGAL, and transforming growth factor- β 1 (TGF- β 1), together with conventional renal biomarkers (urine albumin level, serum creatinine concentration, and serum cystatin C-GFR), to detect nephropathy early in patients with sickle cell anemia. Only KIM-1 and NAG showed a strong correlation with albuminuria; other markers did not show any association with albuminuria.²⁶¹

More recently, plasma KIM-1 was measured in two distinct cohorts at risk for diabetic kidney disease.²⁶² Plasma KIM-1

was measured in subjects from the Action to Control Cardiovascular Risk in Diabetes (ACCORD) and Veterans Administration NEPHROPathy in Diabetes (VA-NEPHRON-D) cohorts. In the ACCORD cohort, baseline KIM-1 levels were higher in those who went on to develop incident CKD. Similarly, in the NEPHRON-D study, those who went on to develop progressive CKD had higher levels at the start of the study (median IQR, 735 pg/mL [438–1172]) compared with 373 pg/mL (225–628; $P < .001$).²⁶² Those in the highest quartile of KIM-1 concentrations in both studies were more likely to experience adverse renal outcomes compared with those in the lowest quartiles of KIM-1 concentrations. This work expands on the limited data that link plasma KIM-1 with CKD progression.²⁶² Although the exact mechanism whereby KIM-1 enters the plasma circulation in the setting of renal injury remains unclear, these promising studies will undoubtedly lead to further investigation of the link between plasma KIM-1 and CKD.

LIVER-TYPE FATTY ACID BINDING PROTEIN

Urinary fatty acid binding protein 1 (FABP1) has been proposed to be a useful biomarker for early detection of AKI and monitoring of CKD. Also known as “L-type” or “liver-type fatty acid binding protein” (L-FABP), FABP1 was first isolated in the liver as a binding protein for oleic acid and bilirubin. FABP1 binds selectively to free fatty acids and transports them to mitochondria or peroxisomes, where free fatty acids are beta-oxidized and participate in intracellular fatty acid homeostasis. There are several different types of FABP, which are ubiquitously expressed in a variety of tissues. At this time, nine different FABPs have been reported—liver (L), intestinal (I), muscle and heart (H), epidermal (E), ileal (II), myelin (M), adipocyte (A), brain (B), and testis (T). L-FABP is expressed in proximal tubules of the human kidney and localized in the cytoplasm. Increased cytosolic L-FABP in proximal tubular epithelial cells may derive not only from endogenous expression, but also from circulating L-FABP that might be filtered at the glomeruli and reabsorbed by tubular cells.

Susantitaphong and associates²⁶³ published a metaanalysis reporting the performance of L-FABP from 15 prospective cohorts and two case-control studies. Although the authors were only able to meta-analyze seven of the cohort studies, they demonstrated that L-FABP levels were 74.5% sensitive (60.4–84.8) and 77.6% specific (61.5–88.2) for the diagnosis of AKI. Additionally, they demonstrated that the results were more promising for the prediction of in-hospital mortality. They concluded that based on the low quality of many of the studies and the varied clinical settings, L-FABP may be a promising biomarker for the early detection of AKI.²⁶³ In what follows we highlight some of the larger and more recent clinical investigations of L-FABP.

Portilla and colleagues have demonstrated that L-FABP predicts the development of AKI within 4 hours of surgery in children undergoing cardiac surgery.²⁶⁴ Others have attempted to validate this finding in the setting of cardiac surgery, with mixed success.^{242,265,266} Recently, TRIBE-AKI published the results of the largest study investigating L-FABP in the setting of adult cardiac surgery, which demonstrated that after adjusting for a clinical model that consisted of factors known to affect the development of AKI, L-FABP did not correlate with the development of AKI in either their

pediatric ($n = 311$) or adult ($n = 1219$) cohort.²¹² It was demonstrated that although L-FABP levels were statistically higher in adults with AKI compared with those with no AKI, the L-FABP concentration (in ng/mL) at the 0- to 6-hour postoperative time point had an AUC of 0.61 for subsequent AKI, and the performance was only marginally better at the 6- to 12-hour time point. Despite this suboptimal performance in detecting incident postoperative AKI, L-FABP was associated with duration of AKI, even after adjusting for factors known to affect the development of AKI. Both the fourth and fifth quintiles were associated with a longer duration of AKI, with adjusted ORs of 1.77 (1.17–2.67) and 1.92 (1.26–2.93), respectively. In the pediatric cohort, although the fifth quintile of L-FABP concentrations at the earliest postoperative time point (0–6 hours) was significantly associated with the development of an AKI odds ratio of 2.9 (1.2–7.1), this effect disappeared after adjustment to 1.8 (0.7–4.6).²⁴³

On a related note, heart fatty acid binding protein (H-FABP) was also measured in the TRIBE-AKI cohort.²⁶⁷ H-FABP is a protein that is predominantly expressed in the cytosol of the myocardium, but can be found in the distal tubule and detects cardiac injury in less than 1 hour.^{268–270} Log-transformed H-FABP concentrations in the preoperative and immediate postoperative periods were associated with an increased risk of AKI (defined by the AKIN criteria), as well as mortality. In analyses adjusted for factors known to be associated with AKI, preoperative H-FABP values were associated with all stages of AKI (adjusted OR, 2.07 [95% CI, 1.48–2.89]) and mortality 1.67 (1.17–2.37), whereas perioperative values were associated with stage 2 or higher AKI (5.39 [2.87–10.11]) in a similarly adjusted analysis.²⁶⁷ In view of these promising results, we anticipate further investigation of H-FABP.

Siew and colleagues²⁷¹ have reported the performance of L-FABP in 380 critically ill subjects from medical, surgical, trauma, and cardiac ICUs, 130 of whom had AKI defined as AKIN stage 1. L-FABP levels were higher in those with AKI ($P = .003$) and were able to discriminate incident AKI with an AUC of 0.59 (0.52–0.65). Although L-FABP was not able to predict the composite endpoint of death or RRT, L-FABP was able to predict the need for acute RRT in a multivariable analysis (HR, 2.36 [95% CI, 1.30–4.25]). These findings mirror those of Doi and colleagues, who published a prospective single-center observational cohort study examining the performance of L-FABP in 339 mixed ICU patients.²¹⁸ In their study, L-FABP outperformed NGAL, IL-18, NAG, and other biomarkers in the detection of AKI, defined by RIFLE risk. Furthermore L-FABP predicted 14-day mortality with an AUC of 0.90. This study, which followed up a smaller study ($n = 145$) by this same group of investigators, has paved the way for L-FABP to be validated for clinical use in Japan.²⁷²

In a cross-sectional study of general hospitalized patients that included 92 participants with AKI and 68 control subjects (26 healthy volunteers and 42 other hospitalized patients, 29 about to undergo coronary catheterization and 13 patients in the ICU with no AKI), Ferguson and colleagues have demonstrated that urinary levels of L-FABP are significantly higher in those with AKI than in hospitalized control patients without AKI, with an AUC of 0.93 (0.88–0.97), with a sensitivity of 83% and specificity of 90% at a cutoff value of 47.1 ng/mg of creatinine.²⁷³ Nickolas and colleagues²¹⁹ examined L-FABP at the time of ER arrival and found that it only had fair discriminatory power with regard to AKI. In their cohort

of 1635 subjects, L-FABP had an AUC of 0.70 (0.65–0.76); however, there was a clear and significant stepwise increase in L-FABP concentrations across the spectrum of AKI (normal < CKD < prerenal < intrinsic AKI).

Because L-FABP is also expressed by the liver, liver injury can be a potential contributor to increased urinary levels of L-FABP during AKI. However, previous studies in patients with CKD, AKI, and sepsis have shown that serum L-FABP levels do not have an influence on urinary levels and that urinary L-FABP levels are not significantly higher in patients with liver disease than in healthy subjects.^{264,274,275}

Urinary L-FABP levels have been investigated as an early diagnostic and predictive marker for contrast medium-induced nephropathy.^{276,277} In a study of adult patients with normal serum creatinine concentrations who underwent percutaneous coronary intervention, serum NGAL levels rose at 2 and 4 hours, whereas urinary NGAL and urinary L-FABP levels increased significantly after 4 hours and remained elevated up to 48 hours after cardiac catheterization.²⁷⁸ Nakamura and associates have demonstrated that baseline urinary L-FABP levels are significantly higher in patients who developed contrast medium-induced nephropathy after coronary angiography; however, the authors did not evaluate the diagnostic performance of urinary L-FABP in predicting AKI.²⁷⁷

Finally, L-FABP has been investigated in the setting of renal transplantation, DGF, and postoperative transplant function.^{257,279,280} When measured in the deceased donor urine, prior to organ harvesting, or the early postoperative setting, urinary L-FABP was associated with donor AKI and mildly predictive of DGF. However, although L-FABP concentrations were associated with DGF, they did not outperform postoperative urine output for this endpoint,²⁸⁰ and they did not predict long-term graft function over the first posttransplantation year.²⁵⁷

Chronic Kidney Disease

In the past, small studies investigating the excretion of L-FABP in the setting of diabetic nephropathy have been mixed. Some have reported a link between decreased urinary concentrations of L-FABP in the setting of renin-angiotensin-aldosterone blockade and preserved GFR, whereas others found no correlation.^{281,282} More recently, L-FABP has been investigated in established cohorts of CKD patients (e.g., ARIC, CRIC). In a case-control study of 321 patients from the ARIC cohort, 135 with ESRD, urinary L-FABP, normalized to urine creatinine, was not associated with an increased risk of ESRD.²⁵³ These results are slightly different from when L-FABP was investigated in the CRIC cohort. Using Cox proportional hazard models, L-FABP was assessed in 2466 members of the cohort, with an average of a 3.8-year follow-up.²⁵⁴ Although several quintiles were associated with progression of CKD (50% drop in eGFR or development of ESRD; fifth quintile unadjusted HR of 10.67 [7.46–15.25]), this effect was entirely attenuated when adjusting for factors known to affect the development and progression of CKD. As such, it appears that L-FABP is not an ideal biomarker for the detection or progression of CKD.

MONOCYTE CHEMOATTRACTANT PROTEIN-1

Monocyte chemoattractant protein-1 (MCP-1) is a chemotactic protein secreted by a variety of cells that attracts blood

monocytes and tissue macrophages through interaction with the cell surface receptor CCR2 (chemokine C-C motif receptor 2).^{283,284} Induction of MCP-1 at the transcript or protein level has been demonstrated in a variety of human cell types, including fibroblasts, endothelial cells, peripheral blood mononuclear cells, and epithelial cells, on proinflammatory stimuli.^{284–288} Kidney cells also produce MCP-1 in response to proinflammatory cytokines, including TNF- α and IL-1 β .²⁸⁹

Acute Kidney Injury

Plasma MCP-1 was measured in 972 participants of the TRIBE AKI cohort to determine its association with postoperative AKI and mortality.²⁹⁰ In this subset of TRIBE patients, 34% of whom had AKI ($n = 329$), MCP-1 levels were significantly higher in those who developed AKI or died (12%; $n = 119$) in the median 2.9-year follow-up period. Preoperative MCP-1 levels in the highest tertile were associated with a 43% increased risk of AKI compared with those in the lowest preoperative tertile; however, this effect was not significant when looking only at an outcome of severe AKI. This analysis was likely limited by the low number of patients who developed severe AKI (stage 2 or 3 AKI, $n = 45$; 5%). Those in the highest preoperative tertile were also at increased risk of death, with an adjusted HR of 1.95 (1.09–3.49).²⁹⁰ Although this TRIBE study represents the largest study to date of the role of MCP-1 as a biomarker of AKI, others have investigated it in other settings, including DGF following kidney transplantation.²⁹¹ Given these limited data, as well as the strength of its role in CKD, we anticipate that MCP-1 will continue to be investigated as a biomarker of AKI over the next several years. Munshi and associates have demonstrated that urinary levels of both MCP-1 transcripts and protein were elevated in patients with AKI, as well as in experimental models of AKI; however, these data have yet to be replicated in other AKI cohorts.²⁹²

Chronic Kidney Disease

The expression of MCP-1 is induced in kidney diseases with significant inflammation, such as diabetic nephropathy and other glomerulonephropathies.^{293–295} In particular, podocytes and tubular cells produce MCP-1 in response to high levels of glucose and advanced glycosylation end products.²⁹⁶ Furthermore, urine levels of MCP-1 are significantly elevated in patients with diabetic nephropathy, and its levels correlate significantly with albuminuria and NAG levels in humans, as well as in experimental diabetic nephropathy.^{297–300} In a prospective observational study of patients with diabetic nephropathy, urinary levels of connective tissue growth factor (CTGF) were elevated in patients with microalbuminuria and macroalbuminuria, but the urinary MCP-1 level was elevated only in those with macroalbuminuria.³⁰¹ Urinary CTGF levels correlated with progression to macroalbuminuria, whereas urinary MCP-1 levels (but not CTGF levels) correlated with the subsequent rate of eGFR decline (at a median follow-up of 6 years). The authors concluded that increased urinary CTGF concentration is associated with early progression of diabetic nephropathy, whereas MCP-1 level is associated with later-stage disease.³⁰¹ The independent association of urine MCP-1 with the risk of CKD progression has been confirmed by others.^{302,303} Elevated levels of urinary MCP-1 were reported in patients with lupus nephritis, and the

presence of MCP-1 in urine reflected its intrarenal expression.^{304,305} Serum concentrations of MCP-1 were also shown to be elevated in patients with diabetic nephropathy and lupus nephritis, but the serum levels did not correlate with disease progression.^{304–306} Moreover, the lack of correlation between urinary and serum MCP-1 levels suggests that urinary MCP-1 is the result of local production of MCP-1 by the kidney rather than simply filtration of serum MCP-1. More recently, Vianna and colleagues demonstrated that plasma and urinary levels of MCP-1 are elevated in pediatric CKD (from either glomerular disease or congenital anomalies) but were not correlated with each other. MCP-1 levels were significantly higher than in those without CKD, and there were differences in MCP-1 concentration, depending on the cause of the CKD.²⁹⁵

NEUTROPHIL GELATINASE-ASSOCIATED LIPOCALIN

NGAL, also known as “lipocalin 2” or “lcn2,” is one of the biomarkers of AKI that has been studied extensively. NGAL has many characteristics required for a good biomarker of AKI compared with serum creatinine measurement or urine output.³⁰⁷ It is a 25-kDa protein with 178 amino acids belonging to the lipocalin superfamily. Lipocalins are extracellular proteins with diverse functions involving transport of hydrophilic substances through membranes, thereby maintaining cell homeostasis.³⁰⁸ NGAL is a glycoprotein bound to matrix metalloproteinase-9 in human neutrophils. It is expressed in various tissues in the body, such as salivary glands, prostate, uterus, trachea, lung, stomach, and kidney,³⁰⁹ and its expression is markedly induced in injured epithelial cells, including those in the kidney, colon, liver, and lung.

Transcriptome profiling studies in rodent models have identified NGAL as one of the most upregulated genes in the kidney very early after tubular injury.^{310,311} Mishra and associates have demonstrated that NGAL levels are significantly elevated within 2 hours after injury in mouse models of renal ischemia–reperfusion.³¹² In addition, urinary NGAL was detectable after 1 day of cisplatin administration, which suggests its sensitivity in other models of tubular injury.³¹²

Acute Kidney Injury

Many clinical studies have followed these important observations in animals. Mishra and associates first demonstrated the value of NGAL as a clinical marker in a prospective study of 71 children undergoing CPB.³¹³ In this study, both serum and urinary NGAL levels were upregulated within 2 hours in patients who developed AKI. A cutoff value of 50 $\mu\text{g/L}$ was 100% sensitive and 98% specific in predicting AKI. Following this seminal paper, several other papers investigating NGAL in the setting of cardiac surgery were published, with several demonstrating that both urine and serum NGAL levels were able to predict AKI earlier than serum creatinine, as well as correlate with AKI severity.*

However, no paper was able to replicate the near-perfect results published in this first study. Given the wealth of studies that have reported on NGAL over the past decade, we have chosen to highlight the larger and multicenter trials that studied NGAL.

*References 44, 45, 108, 109, 210, 211, 213, 242, 244, and 314–319.

Urine Neutrophil Gelatinase-Associated Lipocalin

In the setting of the ER, urine NGAL was originally shown to perform well in Nickolas and colleagues⁶³ first study in 635 patients, demonstrating that a cutoff of 130 $\mu\text{g/g}$ of creatinine had a sensitivity of 90% and specificity of 99.5% for AKI, defined as RIFLE risk.³²⁰ In this single-center prospective study, urine NGAL also predicted the future need for nephrology consult, admission to the ICU, and need for RRT. In a follow-up multicenter study of 1635 subjects, urine NGAL had an AUC of 0.81 (0.76–0.86) for the prediction of AKI (RIFLE risk) and had an NRI of 26.1% while demonstrating the ability to improve the classification of both AKI events and nonevents. Additionally, urine NGAL values were significantly different and increased in a step-wise fashion in those with no AKI < CKD < prerenal AKI < intrinsic AKI.

Lieske and colleagues measured urine NGAL in 363 ER patients and determined that NGAL has an AUC of 0.70 for the detection of AKIN AKI while providing only modest sensitivity (65%) and specificity (65%).^{320a} In addition to demonstrating that NGAL levels increased with the severity of AKI, they demonstrated that pyuria and urinary white blood cells were associated with increased urinary NGAL levels. Urine NGAL has been studied less extensively in the pediatric ER but has demonstrated similar potential in a smaller study ($n = 252$), with a lower AKI incidence rate ($n = 18$; 7.1%).²⁴⁰

Urine NGAL has also been studied in the setting of critical illness by Siew and colleagues, who demonstrated that urine NGAL was able to predict AKI within the first 24 (AUC = 0.71) and 48 (AUC = 0.64) hours of ICU admission.³²¹ In this single-center prospective study of 451 critically ill adults, urine NGAL was independently associated with the development of AKI, even after adjusting for factors known to be correlated with the development of AKI (including severity of illness and sepsis). NGAL performed similarly when measured in a post hoc analysis of the EARLY ARF trial. Data from this prospective observational study performed in two general (mixed) ICUs in New Zealand demonstrated that urine NGAL could modestly predict the development of AKI (AKIN stage 1; AUC, 0.66) but was also able to forecast the need for RRT (AUC, 0.79) and death within the first 7 ICU days (AUC 0.66; $P < .001$ for all three). Additionally, urine NGAL performed better in predicting AKI on ICU arrival in those with higher baseline eGFRs (AUC = 0.70 for those with an eGFR from 90–120 mL/min vs. an AUC of 0.64 for those with an eGFR < 60 mL/min).²¹⁶ This improved ability to detect the future development of AKI in those with a higher eGFR has been demonstrated by others in the setting of cardiac surgery.^{44,45} In a separate post hoc analysis of the EARLY ARF study, urine NGAL values again demonstrated a significant increase across the spectrum of AKI, with concentrations being lowest in those without AKI and values for those with transient AKI that lasted less than 48 hours being in between those with no AKI and those with AKI that lasted longer than 48 hours.²¹⁷ In the discovery phase of the multicenter prospective observational SAPPHIRE trial, with 522 participants, urine NGAL had an AUC of 0.66 (95% CI, 0.60–0.71) for the development of RIFLE injury or failure within the first 36 hours of study enrollment; this increased to 0.71 (0.66–0.76) when looking at RIFLE injury or failure within the first 12 hours.³²²

In the setting of cardiac surgery, urine NGAL has provided similar mixed results. In the TRIBE-AKI adult cohort, the highest quintile of urine NGAL at 0 to 6 hours following surgery was associated with an increased risk of AKI (doubling of serum creatinine or need for RRT); however, this effect was no longer significant after adjusting for factors known to contribute to AKI risk.²¹⁰ In addition to having an AUC of 0.67 for the detection of AKI in this cohort of 1219 adults, urine NGAL levels were significantly associated with the composite endpoint of inpatient mortality or receipt of RRT, as well as length of ICU stay and length of hospitalization. All quintiles of urine NGAL were associated with duration of AKI in unadjusted analyses; however, this effect was no longer significant for all but the fourth quintile after adjusting for factors known to affect the development of AKI.²¹² Urine NGAL did not display the ability to detect AKI progression in the 380 adults who developed at least AKIN stage 1. Although those subjects in the fifth quintile of urine NGAL at the time of serum creatinine increase were at increased risk for more progressive AKI (e.g., going from AKIN stage 1 to AKIN stage 3), this effect was no longer significant in the adjusted analysis.²¹³ These results were in contrast to those of the pediatric cohort ($n = 311$), in which those in the fifth quintile of urine NGAL remained at a significantly increased risk of developing AKI (doubling of serum creatinine or need for RRT), even after adjusting for the clinical model (OR 4.1; 95% CI, 1.0–16.3). Additionally, urine NGAL levels correlated with the length of mechanical ventilation, ICU stay, and hospitalization.²¹¹

In a separate secondary analysis looking at the long-term mortality of the adult TRIBE cohort in patients with AKI ($n = 407$), those subjects in the third tertile of urine NGAL were at increased risk of death during the median 3.0-year follow-up. The adjusted HR for this group (compared with the first tertile) was 2.52 (95% CI, 1.86–3.42). A similar effect was not seen in the third tertile of those without AKI ($n = 792$), HR 0.90 (95% CI, 0.50–1.63).²¹⁴

Urine NGAL has been investigated in the setting of DGF following renal transplantation, with several studies linking elevated urine NGAL levels in the posttransplantation setting with the development of DGF.^{241,323} Reese and colleagues²⁵⁷ shifted this paradigm slightly in measuring urine NGAL in the deceased donor urine rather than in the posttransplantation setting. They demonstrated that elevated urinary NGAL was associated the development of recipient DGF; the third tertile of NGAL had a relative risk of 1.21 (1.02–1.45) compared with the lowest tertile. Elevated deceased donor urine NGAL levels were weakly associated with slightly lower eGFR at 6 months posttransplantation in those patients who did not develop DGF. As such, although urine NGAL did correlate with donor-based AKI, it did not provide meaningful value in predicting short- and long-term outcomes posttransplantation.²⁵⁷

Urine NGAL has been investigated in several other smaller studies in more niche cohorts and has demonstrated some promise in detecting AKI and other adverse patient outcomes, including critically ill neonates,³²⁴ individuals with cirrhosis or hepatorenal syndrome,^{325,326} and 1-year graft survival following transplantation.²²³ However, these trials and others require validation in larger and multicenter investigations.

Plasma Neutrophil Gelatinase-Associated Lipocalin

Plasma NGAL has been examined in many of the same studies with urine NGAL, including in the settings of the ER and ICU and following cardiac surgery. In the ER, Di Somma and colleagues³²⁷ have demonstrated that the plasma NGAL drawn at the time of ER arrival had an AUC of 0.80 for the future development of AKI. In this multicenter prospective cohort study, the AUC improved to 0.90 when ER physician clinical judgment was added to plasma NGAL. This combination of physician clinical judgment and NGAL outperformed either physician judgment and serum creatinine alone, leading to a significant NRI of 32.4%.

In the setting of critical illness, plasma NGAL was measured as part of a post hoc analysis of the multicenter EARLY ARF study ($n = 528$), in which it had an AUC of 0.74 (0.69–0.79) for the development of AKIN stage 1 AKI ($n = 147$) during subsequent ICU stay. This study defined functional AKI based on the AKIN criteria but also defined structural AKI in terms of urine NGAL concentrations. Plasma NGAL performed even better (AUC = 0.79) at predicting urine NGAL-defined structural AKI ($n = 213$). In addition to strong associations with creatinine and urine NGAL-based definitions of AKI, plasma NGAL was also associated with the need for RRT ($n = 19$) but not with inpatient mortality ($n = 53$).³²⁸ In the SAPPHERE trial, which also examined the performance of plasma NGAL in the setting of ICU-associated AKI, NGAL had an AUC of 0.64 (0.58–0.70) for the detection of RIFLE injury or failure within the first 12 hours of study enrollment; this significant ability to forecast more severe forms of AKI did not dramatically change when assessing the ability to detect the same level of AKI over the first 36 hours (AUC, 0.64 [95% CI, 0.58–0.71]).³²²

It has been postulated that NGAL's performance in the setting of critical illness is likely attenuated in part due to the preponderance of sepsis-related AKI in the ICU, and that NGAL levels are inherently higher in those with sepsis because it is derived in part from neutrophils. De Geus and colleagues³⁹ published a prospective observational cohort study of 663 ICU admissions in which plasma NGAL levels were measured four times during the first 24 hours. They demonstrated that plasma NGAL levels were significantly higher in those with sepsis compared with those without sepsis, and that when the cohort was stratified according to the presence of sepsis ($n = 80$; 12% of the cohort), plasma NGAL was able to detect AKI remarkably well in both cohorts (AUC of 0.76 in those with sepsis vs. 0.78 in those without sepsis). These data corroborate the work of Katagiri and associates, which demonstrated in a prospective observational study of 139 critically ill patients that plasma NGAL levels are highest in those with sepsis associated with AKI > nonsepsis AKI > no AKI.³³⁰ Future investigations of plasma NGAL will need to take this association with sepsis into account as we begin to construct normal ranges, as well as clinically validated cutoffs to use NGAL in interventional trials to treat those in the early stages of AKI.^{331,332}

In addition to investigations in the setting of critical illness, plasma NGAL has been investigated in the setting of acute heart failure.³³³ In a multicenter, prospective, observational cohort study of 927 patients with acute decompensated heart failure requiring intravenous diuretics, Maisel and colleagues evaluated plasma NGAL's ability to predict worsening of renal function, defined as a 0.5 mg/dL or $\geq 50\%$ increase in serum

creatinine level from baseline or the need for RRT.³³³ In their study, 72 patients (7.8%) developed the primary end-point with neither first or peak NGAL providing a significantly higher AUC compared with serum creatinine alone (0.656, 0.647, and 0.652 respectively). Similarly, plasma NGAL was predictive of in-hospital adverse events (AUC of 0.691) but did not significantly outperform serum creatinine (AUC, 0.686). This large-scale investigation does not support the use of plasma NGAL for the detection of AKI in the setting of decompensated heart failure.

Despite these disappointing findings in the setting of heart failure, plasma NGAL has been shown to predict recovery from AKI in preliminary studies. In 181 patients with community-acquired pneumonia and at least RIFLE failure AKI, plasma NGAL measures on the first day that met failure criteria were able to predict the failure of recovery of renal function. Individuals with high plasma NGAL levels were less likely to recover, with an AUC of 0.74. However, this was not significantly different from a clinical model consisting of age, serum creatinine level, and severity of illness scores.³³⁴ This potential ability to detect nonrecovery of AKI is yet another aspect of NGAL and other biomarkers that require further investigation.

Plasma NGAL has also been extensively studied in the setting of cardiac surgery. In the TRIBE-AKI adult cohort ($n = 1219$), plasma NGAL levels were significantly higher in those who developed AKI (doubling of serum creatinine or RRT) in the early postoperative period. Those in the fifth quintile of plasma NGAL at the 0- to 6-hour time point (>293 ng/mL) were at a 7.8-fold increased risk of developing AKI compared with those in the first quintile (<105 pg/mL). This effect was attenuated after adjusting for factors known to associate with AKI but remained significant (OR, 5.0 [95% CI, 1.6–15.3]), although it was no longer significant after adjusting for serum creatinine.²¹⁰ Additionally, plasma NGAL was significantly associated with increased length of ICU and hospital stay, as well as a composite of in-hospital death or dialysis. In this same adult cohort, plasma NGAL, measured at the time of a clinical AKI and serum creatinine increase, demonstrated a remarkable ability to detect those individuals with progressive AKI (e.g., going from AKIN stage 1 to stage 2 or 3; $n = 380$). After adjusting for the clinical model, the fifth quintile of plasma NGAL (>322 ng/mL) was nearly eight times more likely to develop progressive AKI (OR 7.72 [95% CI, 2.65–22.49]) compared with those in the first two quintiles. Plasma NGAL displayed the ability to improve the reclassification of both those with and without progressive AKI (events and nonevents), providing a category-free NRI of 0.69 ($P < .0001$).²¹³ The results from the TRIBE-AKI pediatric cohort were less promising, with plasma NGAL not displaying the ability to predict severe AKI (doubling of serum creatinine or need for RRT) in the early postoperative period. However, the fifth quintile of NGAL (>259 ng/mL) measured within the first 6 postoperative hours was significantly associated with the development of RIFLE risk AKI, with an adjusted OR of 2.3 (95% CI, 1.0–5.5),²¹¹ although falling well short of the near-perfect performance published in the original Mishra paper.³³⁵

Outside of these large trials, there have been several smaller trials that investigated NGAL in a variety of AKI settings, so much so that Haase and colleagues conducted a pooled prospective study ($n = 2322$; 1452 with cardiac surgery and 870

with critical illness) that designated subjects as NGAL(+) or NGAL(−) and creatinine(+) or creatinine(−); creatinine(+) was defined as RIFLE-R.⁴¹ After analyzing NGAL data from 10 separate prospective observational studies, they demonstrated that individuals who were NGAL(+) but creatinine(−) needed acute dialysis 16 times more often than those who were NGAL(−) creatinine(−) (OR, 16.4 [95% CI, 3.6–76.9]; $P < .001$). The study also demonstrated an incremental increase in ICU stay, hospital stay and mortality among the four study groups as follows: NGAL(−) creatinine(−) < NGAL(+) creatinine(−) < NGAL(−) creatinine(+) < NGAL(+) creatinine(+).

The function of NGAL as a diagnostic marker of contrast medium-induced nephropathy has also been evaluated. In a prospective study of 91 children undergoing coronary angiography, both urine and plasma NGAL levels were found to be significantly increased within 2 hours of contrast medium administration in the group that developed contrast-induced nephropathy but not in the control group. By comparison, AKI detection using increases in serum creatinine concentration was possible only 6 to 24 hours after contrast agent administration. When a cutoff value of 100 ng/mL was used, both urine and serum NGAL levels at 2 hours predicted contrast medium-induced nephropathy, with AUCs of 0.91 and 0.92, respectively.³³⁶ In several studies of adults undergoing procedures requiring contrast, an early increase in both urine (4-hour) and plasma (2-hour) NGAL levels was documented, compared with a much later increase in plasma cystatin C levels; this provides support for the use of NGAL as an early biomarker for contrast medium-induced nephropathy.^{224,337} A meta-analysis has revealed an overall AUC of 0.89 for the prediction of AKI when NGAL is measured within 6 hours after contrast agent administration when AKI is defined as a 25% or greater increase in serum creatinine concentration.³³⁸

The origin of plasma and urinary NGAL after AKI requires further clarification. Gene expression and transgenic animal studies have demonstrated an upregulation of NGAL in the distal nephron segments, specifically in the thick ascending limb of Henle and the collecting ducts; however, most of the injury in AKI occurs in the proximal tubules.^{339,340} On the other hand, the source of plasma NGAL in AKI is not well defined. For example, in animal studies, direct ipsilateral renal vein sampling after unilateral ischemia indicates that NGAL synthesized in the kidney does not enter the circulation.³⁴⁰ The increase in plasma NGAL observed in AKI may derive from the fact that NGAL is an acute-phase reactant and may be released from neutrophils, macrophages, and other immune cells. Yndestad and colleagues have reported strong immunostaining for NGAL in cardiomyocytes within the failing myocardium in studies of experimental and clinical heart failure.³⁴¹ Furthermore, any impairment in GFR resulting from AKI would be expected to decrease renal clearance of NGAL, with subsequent accumulation in the systemic circulation. However, the contribution of these mechanisms to the rise in plasma NGAL concentration after AKI has yet to be investigated.

NGAL levels are also influenced by various medical conditions, such as CKD, hypertension, anemia, systemic infections, hypoxia, inflammatory conditions, and cancer, which makes it relatively less specific for kidney injury.³⁴² Additionally, there is some evidence to suggest that NGAL concentrations degrade over time, with levels decreasing by

nearly 50% within the first 6 months of storage at -80°C . These degradation issues also affect other biomarkers (including *N*-acetyl- β -D-glucosaminidase and KIM-1); their impact on clinical results remains unclear because this remains an area of continued investigation.³⁴³ Nevertheless, NGAL represents a very promising candidate as a biomarker for the early diagnosis of AKI and potential prediction of outcome.

Chronic Kidney Disease

In addition to extensive investigation in the setting of AKI, NGAL has been increasingly investigated in the setting of CKD. Some of this work was inspired by animal data that demonstrated that NGAL, like KIM-1, was highly upregulated by the persistent inflammation and late immune response following AKI and potentially contributes to the development of post-AKI CKD.²⁴⁸ Moving this concept into humans, Nickolas and colleagues have reported on the correlation of NGAL with histologic changes on native kidney biopsies of subjects with CKD. They demonstrated that NGAL levels were inversely correlated with eGFR and directly correlated with both interstitial fibrosis and tubular atrophy.³²⁰

Over the past several years, there have been several investigations into the ability of urine NGAL to detect incident and progressive CKD, using previously collected cohorts, such as ARIC, MESA, and CRIC. In an original case-control substudy of the ARIC cohort ($n = 286$), urinary NGAL did not initially correlate with baseline eGFR. However, the fourth quartile of urine NGAL was at more than a twofold increased risk of developing incident stage 3 CKD during the follow-up period. It should be noted, however, that this effect was attenuated after adjusting for urine creatinine and urine albumin.²⁵² In a subsequent investigation of ARIC participants, 135 patients with ESRD and 186 matched controls (gender, race, baseline renal function, and diabetes status), there was no association of urine NGAL (normalized to urine creatinine) with the development of ESRD over time.²⁵³

Analysis of data from the MESA cohort was unable to find an association between urine NGAL levels and the future development of incident CKD stage 3. In a 1:1 nested case-control study, NGAL levels were not associated with the development of CKD stage 3 or a decrease in eGFR of >3 mL/min per year over a 5-year follow-up period.²⁴⁹

Finally, several analyses from CRIC have investigated the role of urine NGAL in predicting CKD and its progression. Liu and colleagues have demonstrated that there is a strong association between baseline urine NGAL and the risk of CKD progression (defined as a 50% reduction of MDRD-calculated eGFR or development of ESRD) over the mean follow-up of 3.2 years.³⁴⁴ However, although this effect was significant in an unadjusted analysis, urine NGAL offered no improved prediction after adjusting for baseline age, race, eGFR, proteinuria, diabetes, and other factors known to affect CKD progression (C-statistic of 0.847 for both). Thus, in this cohort of 3386 individuals, urine NGAL offered no improvement in predicting CKD outcomes compared with more traditional markers. More recently, in a subset of 2466 CRIC enrollees with slightly longer follow-up (mean, 3.8 years) Hsu and colleagues demonstrated that the higher four quintiles of urine NGAL (normalized to urine creatinine) were associated with CKD progression. However, after adjusting for factors known to affect CKD progression, this effect was no longer significant in all quintiles.²⁵⁴

These findings from ARIC, MESA, and CRIC, which demonstrate an association with CKD progression that does not persist in adjusted models, are in direct contrast to a recent prospective observational cohort of 158 white patients with baseline CKD stage 3 or 4. This study demonstrated that urine NGAL (adjusted for urine creatinine) was associated with CKD progression; 40 patients reached the primary endpoint of all-cause mortality or need for RRT during the 2-year follow-up. The baseline urine NGAL was associated with this composite primary endpoint, with every increase of urine NGAL of 5 $\mu\text{g}/\text{mmol}$ being associated with a 27% increased risk of death or RRT.³⁴⁵ These findings are similar to those of Bolignano and colleagues, who performed a prospective observational study of 96 subjects with CKD, with a median follow-up of 18.5 months.³⁴⁶ They demonstrated that both urine and serum NGAL were associated with a composite endpoint of either doubling of baseline serum creatinine or the development of ESRD. Conversely, in a 4-year follow-up study of 78 patients with type 1 diabetes conducted to evaluate the potential of urinary NGAL level to predict progression to diabetic nephropathy, NGAL levels were not associated with decline in GFR or development of ESRD and death after adjustment for known progression promoters.³⁴⁷

N-ACETYL- β -D-GLUCOSAMINIDASE

NAG is a lysosomal brush border enzyme that resides in the microvilli of tubular epithelial cells. Damage to these cells results in shedding of this enzyme into the urine. NAG has a high molecular weight of 130 kDa, and hence plasma NAG is not filtered by the glomeruli. Its excretion into urine correlates with tubular lysosomal activity. Increased urinary concentrations of NAG have been found in patients with AKI, chronic glomerular disease, diabetic nephropathy, exposure to nephrotoxic drugs, delayed renal allograft function, environmental exposure, contrast medium-induced nephropathy, or sepsis and following CPB.^{239,348–354} In a prospective study involving 201 hospitalized patients with AKI, patients with higher concentrations of urinary NAG and KIM-1 were more likely to die or require dialysis. This study suggests the utility of NAG in combination with KIM-1 in predicting adverse clinical outcomes in patients with AKI.²³⁹ Urinary NAG concentrations were significantly higher in patients with contrast medium-induced nephropathy than in patients without such nephropathy within 24 hours after the administration of a contrast agent.³⁵³

Similarly, in a two-center Japanese study of 77 patients undergoing cardiac surgery, NAG levels were elevated in those who developed postoperative AKI.²⁶⁵ In this study, biomarker performance significantly improved when NAG was combined with L-FABP (an AUC improvement from 0.75 to 0.81). This same group published a single-center study investigating the performance of NAG in predicting the development of AKI (RIFLE) in a mixed medical-surgical ICU. NAG did not perform as well in this cohort of 339 subjects, providing an AUC of 0.62 for the development of RIFLE risk.²¹⁸ In a cohort of 635 ER patients, a NAG value over 1.0 units/g had an AUC of 0.71 (95% CI, 0.62–0.81) for the development of AKI during the subsequent hospital admission. However, this effect was attenuated in a multivariable analysis that included other novel and traditional biomarkers of AKI (e.g., creatinine, BUN, NGAL).¹⁷⁶

In the setting of CKD, a study of patients with type 1 diabetes and nephropathy, Vaidya and colleagues have shown that lower levels of urinary KIM-1 and NAG are associated with the regression of microalbuminuria.²⁵⁹ Similarly, in a nested case-control study from the Diabetes Control and Compliance Trial, baseline NAG concentrations were shown to predict micro- and macroalbuminuria.³⁵⁵

NAG has been investigated for its ability to detect CKD progression in a few cohorts. In the CRIC cohort, NAG was associated with CKD progression in unadjusted analyses, with the fifth quintile having an HR of 15.16 (10.17–22.59).²⁵⁴ However, after adjusting for age, gender, race, and albuminuria, this association was completely attenuated. Additionally, the ability of NAG to predict CKD progression was analyzed in 149 patients with chronic heart failure over 5-year follow-up.²⁵⁰ Those with a urine NAG level above the median were nearly four times as likely (OR, 3.92; $P = .001$) to have their CKD progress (defined as $\geq 25\%$ decrease in eGFR). NAG was also shown to be an independent predictor of all-cause mortality.²⁵⁰

There are some limitations in the use of NAG as a marker of kidney injury. Inhibition of NAG enzyme activity has been reported in the presence of metal ions and at higher urea concentrations in the urine. Moreover, increased urinary levels of NAG have been reported in several nonrenal diseases, including rheumatoid arthritis and hyperthyroidism, as well as in conditions with increased lysosomal activity without cellular damage.^{356,357} Because of concerns about its specificity, the clinical utility of NAG as a biomarker has been limited.

PROTEINURIA

In a healthy person, urinary protein excretion is less than 150 mg/day and consists mainly of filtered plasma proteins (60%) and tubular Tamm-Horsfall protein (40%).^{358,359} Proteinuria can result from at least three different pathophysiologic mechanisms, including glomerular (increased permeability of glomerular filtration barrier to protein due to glomerulopathy, raised glomerular capillary hydrostatic pressure, or altered glomerular filtration coefficient), overflow (due to increased production of low-molecular-weight plasma proteins, such as immunoglobulin light chains in myeloma), and tubular processes (decreased tubular absorption of filtered proteins or increased production of tubular proteins by damaged tubules). Proteinuria mechanisms and consequences are discussed in Chapter 30. Proteinuria is diagnosed when the total urinary protein is greater than 300 mg/24 hours. Methods for detecting and monitoring proteinuria are discussed in Chapter 23.

Some reports have highlighted the diagnostic power of total protein for AKI in various drug-induced nephrotoxicities, including cisplatin and nonsteroidal antiinflammatory drugs.^{360,361} A low eGFR is a known risk factor for AKI, and the utility of proteinuria in combination with the eGFR to predict the risk of this disease is now being investigated.³⁶² In a large cohort of nearly 1 million adult Canadians, James and colleagues demonstrated an independent association among eGFR, proteinuria, and incidence of AKI.³⁶³ This group reported that patients with normal eGFR levels ($\geq 60 \text{ mL}/\text{min}/1.73 \text{ m}^2$) and mild proteinuria (urine dipstick, trace to 1+) have 2.5 times more risk of admission to a hospital with AKI than patients with no proteinuria. The risk was increased to 4.4-fold when they included patients with heavy

proteinuria (urine dipstick $\geq 2+$). Adjusted rates of admission with AKI and kidney injury requiring dialysis remained high in patients with heavy dipstick proteinuria, independent of the eGFR.¹⁸ These findings confirm previous reports suggesting that the eGFR and proteinuria are potent risk factors for subsequent AKI.^{364,365}

ALBUMINURIA

Albuminuria is recognized as one of the most important risk factors for the progression of CKDs. Albumin is a major serum protein with a size slightly larger than the pores of the glomerular filtration membrane, so albuminuria is best known as a biomarker of glomerular dysfunction; its appearance in large amounts in urine represents compromised integrity of the glomerular basement membrane.³⁶⁶ In smaller amounts, however, the presence of albumin in the urine may reflect tubular injury. Albuminuria is classified in the KDIGO classification system as A1 (UAE < 30 mg/day or urine ACR [uACR] < 30 mg/g creatinine), A2 (previously termed “microalbuminuria”; urinary albumin excretion, 30–300 mg/day, or uACR, 30–300 mg/g creatinine) and A3 (previously termed “macroalbuminuria”; urinary albumin excretion > 300 mg/day or uACR > 300 mg/g creatinine). In a number of clinical studies, albuminuria has been shown to be a sensitive biomarker of drug-induced tubular injury.^{367,368} It is routinely used as a marker of kidney damage for making a CKD diagnosis at eGFR levels above 60 mL/min/1.73 m².³⁵⁹ Guidelines of the National Kidney Foundation (NKF) and American Heart Association (AHA) include microalbuminuria, as well as an increase in the urinary total protein excretion, as a risk factor for renal and cardiovascular disease. Both NKF and AHA guidelines suggest measurement of the uACR in an untimed spot urine sample. Ideally, the uACR should be assessed in at least three different samples to decrease intraindividual variation.³⁶⁹ Albuminuria is a continuous risk factor for ESRD and cardiovascular mortality, with no lower limit, even after adjustment for the eGFR and other established risk factors.^{370–372} Urinary albumin has been used as a biomarker for monitoring CKD progression and potential therapeutic efficacy, although the FDA does not accept albuminuria as a surrogate marker. Using microalbuminuria as a marker, Levin and colleagues have demonstrated that N-acetylcysteine may attenuate contrast-induced glomerular and tubular injury.³⁷³

In the past several years, there have been increased investigations of albuminuria as a biomarker for AKI. In a large-scale, collaborative meta-analysis, Grams and colleagues combined data from eight general population cohorts (1,285,049 subjects) and five CKD cohorts (79,519 subjects) to investigate the association of albuminuria and other factors with AKI.³⁷⁴ The primary outcome for this study was hospitalization for AKI. Using Cox proportional hazards models, they demonstrated that the incidence of AKI was higher in those with CKD (2.6%) compared with the general population (1.3%). Additionally, compared with those with an uACR < 5 mg/g, the risk of AKI with uACR > 300 mg/g was 2.73 (2.18–3.43).³⁷⁴ Thus, increased uACR is a strong risk factor for the long-term risk of developing of AKI.

The TRIBE AKI measured preoperative albuminuria in 1159 adult patients, organizing the cohort into clinical risk categories based on the preoperative uACR: 10 mg/g or less (≤ 1.1 mg/mmol), 11 to 29 mg/g (1.2–3.3 mg/mmol), 30 to

299 mg/g (3.4–33.8 mg/mmol), and 300 mg/g or greater (≥ 33.9 mg/mmol). The incidence of AKI, defined as AKIN stage 1, increased across the uACR categories with those in the > 300 mg/g category having a relative risk (RR) of 2.36 (95% CI, 1.85–2.82) compared with the < 10 mg/g group. This association was slightly attenuated after adjusting for variables known to affect proteinuria and AKI, with a RR of 2.21 (95% CI, 1.66–73).³⁷⁵ Similarly, the fourth and fifth quintiles of urine albumin (in mg/L) were associated with increased duration of AKI in unadjusted analyses (fifth quintile, OR 2.83 [1.94–4.12]).²¹² Although this effect disappeared for the fourth quintile after adjusting for factors known to affect the development of AKI, there was only slight attenuation of the fifth quintile (2.21 [1.48–3.30]).²¹² These adult data are in contrast to pediatric TRIBE-AKI data ($n = 294$), which demonstrated no association between the preoperative uACR and the development of postoperative AKI.³⁷⁶ The adult uACR data represent an additional biomarker to aid in cardiac surgery AKI prediction models and supplement other recent data that point to proteinuria and albuminuria serving as biomarkers of AKI in the preoperative and postoperative setting.^{377,378}

In the postoperative setting, the TRIBE-AKI cohort demonstrated that urinary albumin concentrations (mg/L) and dipstick proteinuria values within 6 hours of adult cardiac surgery correlated with the future development of AKI. Compared with the lowest quintile, the highest quintile of albuminuria (mg/L) and highest group of dipstick proteinuria were associated with the greatest risk of AKI (adjusted RR, 2.97 [95% CI, 1.20–6.91]), and adjusted RR of 2.46 (95% CI, 1.16–4.97), respectively]. However, only the postoperative urine albumin concentration (mg/L; not indexed to urine creatinine) was associated with improved risk stratification when added to the clinical model (AUC increased from 0.75 to 0.81; $P = .006$). Urinary albumin (mg/L) in the early postoperative period was also highly predictive of long-term mortality in the TRIBE AKI adult cohort. Specifically, in those with perioperative AKI ($n = 407$), those in the second tertile of albuminuria were at increased risk of death in the 3.0-year follow-up period (adjusted HR 2.28 [95% CI, 1.06–4.88]). Although this effect was further magnified in the third tertile of those with AKI (HR, 2.85 [95% CI, 1.36–5.99]), there was no increased mortality across any of the tertiles of urine albumin in the 792 subjects without perioperative AKI.²¹⁴ Despite its known utility in other settings, a higher early postoperative uACR (mg/g creatinine) was not statistically associated with AKI risk. The poor performance of uACR in the context of adult cardiac surgery may be explained by variations in the urine creatinine excretion within and between individuals, which could be especially prominent when renal function is not in a steady state, and may in part explain why urine albumin concentration (mg/L) outperformed uACR.

In the TRIBE AKI pediatric cohort, perioperative values of the uACR (mg/g), and not urine albumin concentration (mg/L), were found to be predictive of AKI. In children younger than 2 years, an absolute first postoperative uACR ≥ 908 mg/g (103 mg/mmol, highest tertile) predicted the development of AKIN stage 2 or 3 AKI, with an adjusted RR of 3.4 (95% CI, 1.2–9.4) when compared with the first tertile. In children 2 years of age or older, a postoperative uACR ≥ 169 mg/g (19.1 mg/mmol, highest tertile), regardless of

preoperative values, predicted stage 1 AKI after adjusting for clinical factors, such as age, race, gender, and preoperative eGFR, type of cardiac surgery, and adjusted RR of 2.1 (95% CI, 1.1–4.1).³⁷⁶ Although urine albumin concentration and uACR remain established and readily available laboratory tests, the diversity of results when investigating the development of postoperative AKI indicates that further studies are needed before it may be used in clinical practice.

URINARY CYSTATIN C

Urinary cystatin C tracks the function of proximal tubular cells. In healthy individuals, the urinary levels of cystatin C are almost undetectable. Any damage to proximal tubular cells can impede the reabsorption and enhance the urinary excretion of cystatin C. Several clinical studies have sought to understand the potential of urinary cystatin C levels for the prediction of kidney injury and its prognosis. Herget-Rosenthal and associates analyzed data from 85 patients in the ICU who were at high risk of developing AKI and used the RIFLE classification to define AKI.^{379,380} In their studies, the authors reported that the urine cystatin C level detected AKI 1 to 2 days before changes in serum creatinine level, with an AUC of 0.82 and 0.97 on day 2 and day 1, respectively, as well as demonstrating that urine cystatin C serves as a marker of AKI severity correlating with a future need for RRT. Urinary cystatin C-to-creatinine ratios of more than 11.3 mg/mmol were significantly associated with proteinuria. Attempts to validate urine cystatin C as a marker of ICU-associated AKI have provided mixed results. Siew and colleagues have measured urine cystatin C in 380 ICU patients (mixed surgical-medical, trauma, and cardiac) and demonstrated that there is no difference in concentrations between those with and without AKI ($P = .87$).²⁷¹ More encouraging data from the EARLY ARF trial have demonstrated that in a cohort of 529 subjects, urine cystatin C may have limited ability to detect AKI (AUC of 0.67 on ICU arrival), with no significant difference in its ability to detect AKI in those with a GFR above and below 60 mL/min.²¹⁶ Additionally, in a separate post hoc analysis of the same study, urine cystatin C values exhibited a stepwise and significant ($P < .001$) increase when comparing values from those with no AKI < prerenal AKI < intrinsic AKI (defined as AKIN stage 1 > 48 hours).

In comparison with these mixed ICU-associated AKI data, several small studies investigating urine cystatin C in the setting of cardiac surgery have demonstrated promise.^{44,109,317,381} However, these results were not validated in the TRIBE-AKI cohort. In unadjusted analyses of the adult cohort, several quintiles of urine cystatin C were significantly associated with the development of both mild (AKIN stage 1) or severe (doubling of creatinine or need for RRT) AKI at both the 0- to 6-hour and 6- to 12-hour postoperative time points. However, the small associations were completely attenuated after adjusting for the clinical model. Similarly, in the TRIBE pediatric cohort, no quintile remained significantly associated with AKI (mild or severe) in the adjusted analyses.³⁸² Urine cystatin C demonstrated similar results when measured in a cohort of 1635 ER patients, supplying an AUC of 0.65 (95% CI, 0.58–0.72) for the future development of AKI. However, using a multivariable analysis that included traditional (creatinine) and more modern (urine NGAL, KIM-1, IL-18, and LFABP) biomarkers, urine cystatin C was not a significant contributor in predicting the composite outcome of inpatient

RRT or death.²¹⁹ Finally, when investigated in a prospective multicenter observation cohort study of deceased donor kidney transplants, urine cystatin C concentration from the first postoperative day was modestly correlated with 3-month allograft function; the AUC for predicting DGF at the 6-hour postoperative time point was 0.69.³⁸³

A number of studies have reported increased urinary cystatin C levels in patients with proteinuria, which suggests the possibility of tubular damage as a consequence of protein overload.^{384–386} Currently, urinary cystatin C level has several disadvantages as a biomarker, including lack of international standardization and expense of the assay. Although serum cystatin C has been demonstrated as a reliable biomarker of eGFR, one must remember that cystatin C synthesis is increased in smokers, patients with hyperthyroidism, those receiving glucocorticoid therapy, and those with elevated levels of inflammatory markers, such as white blood cell count and C-reactive protein level, and the impact of these factors on urine cystatin C in the setting of AKI has not been fully investigated.^{387,388} Furthermore, several different commercial assays are available to measure cystatin C. Advantages are that the commercially available immunonephelometric assay provides rapid automated measurement of cystatin C, and results are available in minutes.³⁸⁹ In addition, preanalytic factors, such as routine clinical storage conditions and freezing and thawing cycles, and interfering substances, such as bilirubin or triglycerides, do not affect cystatin C measurement.^{389,390}

PROENKEPHALIN

Enkephalins are small endogenous opioid peptides; because they are known to be expressed in the kidney, they have been investigated for the ability to detect both AKI and CKD.³⁹¹ Physiologically, preproenkephalin is the primary gene product of the *PENK* gene, but this propeptide is eventually cleaved, and proenkephalin (pro-ENK) is generated in this process. Given its low molecular weight, it is freely filtered at the glomerulus and, as such, is potentially ideal as a biomarker of glomerular function. Plasma pro-ENK concentrations have been shown to be stable in vitro for 48 hours, unlike other enkephalins, which have proven more difficult to measure (e.g., met-enkephalin).³⁹²

Acute Kidney Injury

Pro-ENK has been investigated in a few AKI clinical settings, including a retrospective observational cohort of patients with suspected sepsis in the emergency department.³⁹³ Given that it is freely filtered, plasma levels of pro-ENK were inversely correlated with creatinine clearance ($r = -0.72$). Additionally, pro-ENK levels were significantly higher in those who eventually developed AKI (defined by RIFLE criteria), compared with those without AKI. Pro-ENK on arrival in the ER had an AUC 0.815 ($P < .001$) for the future development of AKI. Additionally, pro-ENK was modestly associated with 7-day inpatient mortality, with an AUC of 0.69; this significantly outperformed serum creatinine for this endpoint (AUC, 0.61; $P = .045$).³⁹³

Pro-ENK has also been evaluated for its ability to detect AKI following adult cardiac surgery.³⁹⁴ In a single-center cohort of 92 patients, 20 of whom developed postoperative AKI (AKIN definition), preoperative and early postoperative pro-ENK levels were higher in those who went on to develop AKI. Preoperative values provided a C-statistic of 0.68

($P = .013$); early postoperative values gave a C-statistic 0.72 ($P < .001$).³⁹⁴

In a separate investigation, pro-ENK levels were measured in 1908 patients with acute heart failure to investigate its relationship with renal function, as well as short- and long-term mortality.³⁹⁵ Pro-ENK levels were independently associated with an increased risk of worsening of renal function ($n = 264$), defined as 0.3 mg/dL or 50% increase in serum creatinine from the admission value within the first 5 days. In an adjusted analysis, elevated pro-ENK levels on admission were associated with increased odds of developing a subsequent rise in serum creatinine (OR, 1.58 [1.25–2.00]; $P < .0005$). In this multicenter cohort of patients with cardiorenal syndrome and AKI, after multivariable Cox proportional hazards analyses, pro-ENK was independently associated with in-hospital and 1-year mortality. Finally, in the largest investigation of pro-ENK to date, pro-ENK was highly correlated with eGFR and plasma urea levels, with an F-statistic of 296 and 166, respectively ($P < .001$ for both).³⁹⁵

Chronic Kidney Disease

Although pro-ENK appears to be a useful biomarker in the setting of early AKI, its relationship with eGFR requires further evaluation, because limited data have suggested that it may also serve as a biomarker of CKD.³⁹⁶ In a prospective cohort of 2568 patients without CKD (defined as eGFR > 60 mL/min/1.73 m²), plasma pro-ENK was found to be associated with incident CKD. Those in the highest tertile of baseline pro-ENK level had an increased incidence of CKD compared with the lowest tertile (OR 1.51 [1.18 to 1.94]; $P < .001$). Additionally, pro-ENK improved the reclassification of patients for the detection of incident CKD in over 14% of the cohort. Additionally in this same cohort, there was a signal that a minor allele in the pro-ENK gene was also associated with an increased risk of CKD. Although much like the other associations, these need to be validated in separate cohorts.³⁹⁶

URINE TISSUE INHIBITOR METALLOPROTEINASE-2 AND INSULIN-LIKE GROWTH FACTOR-BINDING PROTEIN-7

Urine tissue inhibitor metalloproteinase-2 and insulin-like growth factor-binding protein-7 (TIMP-2 and IGFBP-7) have been shown to serve as biomarkers of AKI in the setting of critical illness. These biomarkers are unique in that they play a role in cell cycle arrest. Both IGFBP-7 (through p523 and p21) and TIMP-2 (through p27) block the effect of cyclin-dependent protein kinase complexes and cause short periods of G1 cell cycle arrest.^{397–399} They were originally discovered as part of a three-center discovery cohort of 522 subjects. These patients had AKI stemming from sepsis, shock, major surgery, and trauma. Over 300 potential markers were evaluated, with TIMP-2 and IGFBP-7 being the two that best predicted the development of KDIGO stage 2 or 3 AKI. This finding was then validated in a prospective international multicenter observational study of 728 subjects.³²² In this validation study, TIMP-2 and IGFBP-7 remained the top two performing biomarkers for the prediction of RIFLE injury or failure within the first 12 to 36 hours of study enrollment, with AUC values of 0.77 and 0.75, respectively (SAPPHIRE study). When these two biomarker values were multiplied together, they demonstrated an improved ability to detect this same endpoint (AUC of 0.80). The paper did not supply

information about the combination of TIMP-2 and IGFBP-7 with other biomarkers of AKI (e.g., NGAL, KIM-1, L-FABP, IL-18).³²² However, these same biomarkers (TIMP-2 and IGFBP-7) were further validated in the OPAL study, where they provided a similar AUC for the detection of impending stage 2 or 3 AKI.⁴⁰⁰ Perhaps more importantly, this study identified cut points for TIMP-2 $\times$ IGFBP-7 to help risk-stratify patients for severe AKI. Patients with a value less than 0.3 were at lowest risk for severe AKI, those with a value between 0.3 and 2.0 had a RR 4.7 (1.5–16.0) times higher, and those with a value greater than 2.0 had a RR 12.0 (4.2–40) times higher than the lowest risk group. These data cut points were further validated in a post hoc analysis of the original 728 SAPPHIRE cohort.⁴⁰⁰

This reproducibility of these results in distinct cohorts led the FDA to clear TIMP-2 $\times$ IGFBP-7 for clinical use in late 2014.^{322,400,401} Additionally, Liu and colleagues have published data demonstrating that biomarker concentrations correlate to the clinical adjudication of a panel of three expert nephrology adjudicators who were blinded to the biomarker values.⁴⁰² Since that time, several post hoc investigations of these and other cohorts have demonstrated that these biomarkers have several strengths. In the 9-month follow-up of the SAPPHIRE cohort, TIMP-2 $\times$ IGFBP-7 levels measured at the time of study enrollment were shown to correlate with a composite endpoint of death or need for RRT.⁴⁰³ There was a stepwise increase in risk of the endpoint across the biomarker cutoff strata (0.3 and 2.0) in those subjects who went on to develop AKI; those with a value greater than 2.0 had an adjusted HR of 2.16 [1.32–3.53] compared with those with values less than 0.3.⁴⁰³ TIMP-2 $\times$ IGFBP-7 has also been shown to reliably forecast impending severe AKI in several subsets of critically ill patients, including those with CKD and congestive heart failure, as well as those undergoing emergent and cardiothoracic surgery.^{404,405} Additionally, others have investigated these biomarkers in other clinical settings, such as pediatric AKI⁴⁰⁶ and DGF following renal transplantation,²⁹¹ and further validation of these markers is required outside the setting of adult critical illness. Given the FDA and European clearance for clinical testing, the American Society of Nephrology's AKI Advisory group and others have published guidance documents on the potential clinical use of TIMP-2 $\times$ IGFBP-7 (Box 27.2).^{407,408}

Furosemide Stress Test

Furosemide and other diuretics have long been used in the setting of AKI to convert oliguric AKI into nonoliguric AKI, but they have seen a resurgence for the ability to ascertain

Box 27.2 TIMP-2 $\times$ IGFBP-7

1. Urinary TIMP-2 $\times$ IGFBP-7 has been FDA-approved for determining which high-risk critically ill patients are at increased risk of severe AKI.
2. Elevated TIMP-2 $\times$ IGFBP-7 levels (>0.3) measured early in the course of an ICU stay can forecast the development of at least stage 2 AKI and have been shown to be associated with posthospitalization outcomes, such as 9-month mortality and the need for RRT.

tubular reserve in the setting of AKI.^{409–411} As a loop diuretic, furosemide (an organic anion) is not filtered by the glomerulus because it is tightly bound to serum proteins (albumin), and it therefore gains access to the tubular urinary space through active secretion in the proximal convoluted tubule via the human organic anion transporter (hOAT) system.^{412,413} In the tubular lumen, furosemide inhibits the ability of the thick ascending limb of the loop of Henle to transport chloride, preventing sodium reabsorption and resulting in natriuresis and increased urine flow and potentially reducing tubular oxygen demand.^{414–418} As a result of these pharmacokinetic factors, the urine output following furosemide dosing has been hypothesized to represent an approach to assess the renal tubular integrity in the setting of AKI.

In 1973, Baek and colleagues studied 15 subjects who did not yet have clinically apparent AKI (as measured by changes in serum creatinine).⁴¹⁰ They subjected them to a one-time furosemide challenge assessing their UOP and free water clearance (C_{H_2O}). Although they did not standardize the furosemide dosing, or report if any of their patients had preexisting CKD, they did report that a C_{H_2O} near zero and a poor urine output response to furosemide signaled that “acute renal failure was imminent.”⁴¹⁰

More recently, Chawla and colleagues have modified and standardized this protocol, renaming it the “furosemide stress test” (FST).⁴¹⁹ They prospectively investigated the ability of the urinary response following the administration of the FST to predict adverse patient outcomes in patients with KDIGO stages 1 and 2 AKI.³⁹ In a cohort of 77 subjects, they reported that the 2-hour urine output response to 1.0 mg/kg FST (in the furosemide-naïve) and 1.5 mg/kg FST (in those with prior loop exposure) was able to forecast the progression to KDIGO stage 3 AKI ($n = 25$; 32.4%), with an ROC AUC (standard error) of 0.87 (0.05). A cutoff of 200 mL at 2 hours provided a sensitivity and specificity of 87.1% and 84.1%, respectively, for progression to stage 3 AKI. This same cutoff provided an AUC (SE) of 0.86 (0.08) for the receipt of inpatient RRT ($n = 11$; 14.2%) and 0.70 (0.09) for inpatient mortality ($n = 16$; 20.7%).⁴¹⁹

In a secondary analysis of this same cohort, FST outperformed plasma and urine NGAL, urinary IL-18, and TIMP-2 and IGFBP-7 for prediction of progression to stage 3, need for RRT, and mortality.⁴²⁰ Interestingly, when the FST data were analyzed in a subset of patients with elevated biomarkers (urine NGAL > 150 ng/mL; $n = 44$ or TIMP-2 $\times$ IGFBP-7 > 0.3; $n = 32$), its ability to detect these same outcomes was further improved. Thus, combining AKI biomarkers with a functional assessment of tubular reserve like the FST may be an informative bedside tool to assist clinicians in the prognostication of early AKI.⁴²⁰ In this study, the FST was not performed in subjects deemed to be hypovolemic, those with obstructive uropathy, or those with a baseline eGFR less than 30 mL/min/1.73 m².⁴¹⁹

Conversely, Van der Voort and colleagues⁴²¹ investigated the ability of a standardized infusion of furosemide to predict renal recovery in those with advanced AKI requiring RRT. In a post hoc analysis, they reported that urine output (median IQR) was higher in patients who recovered their renal function and eventually stopped RRT (654 mL [333–1155] vs. 48 [15–207] mL; $P = .007$). Their version of the FST for prediction of recovery had a diagnostic performance AUC ROC of 0.84. Regardless of the clinical scenario in which it is being

tested, the urinary response to furosemide informs the clinician about the renal reserve present during AKI (progression or recovery). The FST provides a measure of proximal tubule secretory capacity but also assesses thick ascending limb and distal collecting duct function. Thus, the FST serves as an assessment of integrated nephron function and warrants further validation in larger and more diverse cohorts.

GENETIC ASSOCIATIONS WITH ACUTE KIDNEY INJURY RISK

In addition to investigating novel proteins and functional aspects of the nephron, others have focused their attention on the role that genetics play in predisposing patients to an increased risk of AKI and CKD. Discussions of the genetics of chronic kidney disease can be found in Section VI of this edition and are beyond the scope of this chapter. With regard to AKI, there have been several investigations over the past few years reporting associations with several polymorphisms.

Susantitaphong and colleagues have demonstrated that a polymorphism in the promoter region of the TNF- α gene was associated with AKI severity and distant organ dysfunction.⁴²² Using a cohort of 262 hospitalized adults, they demonstrated that after adjustment for race, gender, age, baseline eGFR, sepsis, and the need for RRT, those with the minor allele (rs1800629) had higher peak serum creatinine and higher multiple organ failure scores, compared with the GG genotype. In a cohort of 401 patients with acute lung injury enrolled in the Fluids and Catheter Treatment Trial (FACTT), Bhatraju and colleagues have reported that two minor alleles in the nuclear factor of kappa light-chain polypeptide gene enhancer in B cells inhibitor (NFKBIA) are strongly associated with the development of AKIN criteria for AKI. For rs1050851 and rs2233417, the ORs for any AKI were 2.34 (1.58–3.46; $P = 1.06 \times 10^{-5}$; false discovery rate (FDR) = 0.003) and 2.46 (1.61–3.76; $P = 1.81 \times 10^{-5}$; FDR = 0.003) for each minor allele, respectively.⁴²³ The associations were increased for AKIN stages 2 and 3, with respective ORs of 4.00 (2.10–7.62; $P = 1.05 \times 10^{-5}$; FDR = 0.003) and 4.03 (2.09–7.77; $P = 1.88 \times 10^{-5}$; FDR = 0.003).

In one of the larger investigations around the genetic basis for postoperative AKI, Stafford-Smith and colleagues used a discovery cohort of 873 nonemergent coronary bypass patients and then attempted to replicate their findings in a 380-subject validation cohort.⁴²⁴ Using linear regression, they adjusted for a clinical AKI risk score to test single-nucleotide polymorphism (SNP) association with peak postoperative serum creatinine level. In the discovery cohort, nine SNPs that met statistical significance were detected; two of these, rs13317787 in GRM7|LMCD1-AS1 intergenic region (3p21.6) and rs10262995 in BBS9 (7p14.3), were replicated with significance in the validation cohort. Further investigation of these two regions and meta-analyses found genome-wide significance at the GRM7|LMCD1-AS1 locus and a significantly strong association at BBS9.⁴²⁴ These are two new loci that had not been previously described.

More recently, the largest investigation of the genetic susceptibility to in-hospital AKI was conducted by Parikh and colleagues.⁴²⁵ They performed an exploratory genome-wide association study in 760 cases of AKI and 669 controls (hospitalized patients without AKI) and attempted to validate their findings in an additional 206 cases of AKI and 1406

controls. They assessed 609,508 SNPs and, in their replication analyses, they validated four SNPs with an increased odds of association with the development of AKI. These SNPs included rs62341639 (OR, 0.64 [0.55–0.76]; $P = 2.48 \times 10^{-7}$) and rs62341657 (0.65 [0.55–0.76]; $P = 3.26 \times 10^{-7}$) on chromosome 4 near the APOL1-regulator IRF2, and rs9617814 (0.70 [0.60–0.81]; $P = 3.81 \times 10^{-6}$) and rs10854554 (0.67 [0.57–0.79] $P = 6.53 \times 10^{-7}$) on chromosome 22.⁴²⁵

Prior to these more recent larger studies, a handful of systematic reviews of the genetic predisposition to AKI were published.^{426–428} They concluded that despite several gene polymorphisms with documented associations with the development, severity, and outcome of AKI, definitive conclusions would require more investigation, with replication in new cohort studies.⁴²⁷ We anticipate further investigation of the role of genetics in AKI risk in the years to come.

CHRONIC KIDNEY DISEASE BIOMARKERS

Currently, the eGFR and proteinuria are used as markers of CKD progression because of their widespread availability and ease of performing the tests. Because all forms of CKD are associated with tubulointerstitial injury, markers of tubular injury, including KIM-1, IL-18, NGAL, and L-FABP, have been investigated in CKD and shown to associate with outcomes in CKD due to a variety of causes, as discussed previously. In addition, elevated systemic levels of molecules that have impaired kidney clearance or increased production in CKD (e.g., asymmetric dimethylarginine, fibroblast growth factor 23) as well as chemokines (e.g., monocyte chemoattractant protein-1) and fibrotic markers (e.g., connective tissue growth factor, TGF- β 1, and collagen IV) are discussed here. For more in-depth discussion around specific markers for glomerular diseases, cystic diseases, diabetes, and inherited forms of renal disease, please see the individual chapters covering these topics.

Table 27.7 summarizes the ability of biomarkers to detect clinical endpoints related to CKD in a variety of clinical settings.

PLASMA ASYMMETRIC DIMETHYLARGININE

Nitric oxide is synthesized by oxidation of the terminal guanidine nitrogen of L-arginine by nitric oxide synthase (NOS). This process can be reversibly inhibited by guanidine-substituted analogs of L-arginine, such as in asymmetric dimethylarginine (ADMA).^{429,430} Three types of methylated arginines have been described in vivo—ADMA, N^G-monomethyl-L-arginine, and symmetric dimethylarginine, an inert isomer of ADMA. Of these, ADMA is the major type of endogenously generated methylated arginine that displays inhibitor activity of NOS. However, administration of ADMA to endothelial NOS knockout mice also induces vascular lesions, which suggests that ADMA may have actions independent of nitric oxide and NOS in vivo.⁴³¹ Vallance and associates first reported that plasma levels of ADMA are elevated in patients with renal failure⁴³² and hypothesized that impaired renal clearance of ADMA may account for the rise in plasma levels. This assumption has been challenged by follow-up studies in animal models demonstrating that only a small portion of circulating ADMA is excreted in the urine.⁴³³ Moreover, elevated plasma ADMA levels are also reported in patients with incipient renal disease but normal renal function.⁴³⁴

Elevated plasma levels of ADMA have been reported in patients with a variety of cardiovascular risk factors, such as hypertension, diabetes, and hyperlipidemia.^{435–437} Among these groups, plasma ADMA levels are particularly high in patients with CKD, patients with ESRD undergoing hemodialysis or peritoneal dialysis, and kidney transplant recipients.^{432,438,439} Plasma levels of ADMA are strongly associated with carotid intima to media thickness, left ventricular hypertrophy, cardiovascular complications, and mortality in patients with ESRD.^{440–442} Plasma ADMA levels have been shown to correlate prospectively, in an inverse manner, with CKD progression in those with and without diabetes-related CKD.^{443,444} Similarly, ADMA levels may also be associated with the presence of proteinuria and albuminuria.⁴⁴⁵

Table 27.7 Biomarker Performance in Detecting Chronic Kidney Disease (CKD): Multicenter Studies

Parameter	Diagnosis of CKD	Progression of CKD to ESRD	Cardiovascular and Mortality Risk Assessment in CKD	Progression of HIV-Associated CKD
Urine NGAL	+	–	?	?
Blood NGAL	+	–	?	?
Blood CysC	+	+	+	+
Urine IL-18	–	?	?	+
Urine KIM-1	+	+	?	–
Plasma KIM-1	+	+	?	?
β -Trace protein	–	?	+	?
Urine protein and albumin	+	+	+	+
FGF23	–	+	+	?
TNFR1, R2	+	+	?	?
suPAR	+	+	?	?
EGF	+	+	?	?

CysC, Cystatin C; EGF, epidermal growth factor; ESRD, end-stage renal disease; FGF23, fibroblast growth factor 23; IL-18, interleukin-18; KIM-1, kidney injury molecule-1; NGAL, neutrophil gelatinase-associated lipocalin; suPAR, soluble urokinase-type plasminogen activator receptor; TNFR, tumor necrosis factor receptor; +, data published display the ability to detect this aspect of CKD; –, data published do not display the ability to detect this aspect of CKD; ?, no large multicenter data published on this biomarker or aspect of CKD.

More recently, larger longitudinal studies have been performed linking ADMA with several forms of CKD and CKD outcomes. In a nested case-control study performed in the Genetics of Diabetes Audit and Research Tayside Study (GO-DARTS), ADMA and several other biomarkers were measured to determine their association with CKD progression in patients with type 2 diabetes.⁴⁴⁶ In this study, progression of CKD was defined by a 40% or greater reduction in baseline eGFR over the 3.5-year study period, whereas in controls the eGFR decreased by less than 5% over this same period. ADMA levels were strongly associated with CKD progression in an analysis adjusted for clinical variables known to affect CKD progression, as well as several other biomarkers. Higher ADMA levels were associated with more rapid progression of CKD; for each standard deviation change in ADMA levels there was an increased adjusted odds of 8.36 (3.83–20.4; $P < .001$). This effect was larger than any of the other 14 biomarkers that they measured in this cohort, including KIM-1 (1.93 [1.18–3.27]; $P = .011$) and H-FABP (0.63 [0.38–1.02]; $P = .06$).⁴⁴⁶ In a prospective observational cohort of 1157 patients (663 with diabetic kidney disease, 273 with glomerulonephritis, and 221 with cystic and/or interstitial disease), ADMA levels were measured to determine their association with mortality prior to the initiation of RRT. ADMA levels were associated with an increased risk of mortality in both the diabetic cohort (HR, 1.3 [1.1–1.6]) and the glomerulonephritis cohort (HR, 1.5 [1.3–1.8]).⁴⁴⁷ Despite these emerging data, larger longitudinal studies are needed to demonstrate the ability of ADMA to identify CKD and predict its progression in cohorts with CKD due to multiple causes.

FIBROBLAST GROWTH FACTOR 23

Fibroblast growth factor 23 (FGF23) is a 32-kDa protein consisting of 251 amino acids coded by the FGF gene located on chromosome 12 in the human genome. FGF23 serves as an endocrine hormone that is secreted by osteoblasts and osteoclasts, which binds a FGF receptor and its coreceptor Klotho and stimulates phosphaturia.^{448–450} In addition to promoting phosphaturia, FGF23 also decreases levels of 1,25-dihydroxy vitamin D, as well as parathyroid hormone (PTH).^{451,452} Several cross-sectional studies have demonstrated that FGF23 levels are elevated in both adult and pediatric CKD populations.^{453–455} In a prospective observational study of 3879 subjects in CRIC, FGF23 levels were shown to be increased in individuals with CKD stages 2 to 4, with FGF23 elevation occurring in the absence of abnormalities in serum phosphate and PTH.⁴⁵⁶ As such, FGF23 has emerged as a candidate biomarker to detect CKD and abnormalities of bone mineral metabolism in the absence of changes in serum phosphate and PTH.

Apart from bone, FGF23 is principally expressed in the ventrolateral thalamic nucleus in mice and is also known to be secreted in minimal amounts in the liver, heart, thymus, and lymph node. Maintenance of phosphate homeostasis is carried out by the sodium-dependent phosphate cotransporters NaPi-IIa and NaPi-IIc at the brush border membrane of proximal tubule cells in kidney, and FGF23 has been shown to regulate the activity of these transporters.^{457,458}

FGF23 is increased in CKD and is a prognostic indicator for cardiovascular disease in patients with CKD.⁴⁵⁹ In a recent analysis of 3860 participants of CRIC, FGF23 levels in the highest quartiles (compared with the lowest) were

independently associated with a graded risk of congestive heart failure (HR, 2.98 [95% CI 1.97–4.52]) and atherosclerotic events 1.76 (95% CI, 1.20–2.59), even after adjustment for eGFR, proteinuria, and other traditional cardiovascular risk factors.⁴⁶⁰ Other studies have shown that elevated plasma FGF23 concentrations are associated with cardiovascular events in patients not requiring dialysis, as well as with mortality in patients receiving hemodialysis, with levels in those with ESRD reaching nearly 1000-fold above normal.⁴⁶¹ Interestingly, FGF23 levels have also been shown to decline rapidly in individuals with prompt allograft function in the setting of renal transplantation.^{462,463} It has been reported that serum FGF23 concentrations may be a useful marker for predicting the future development of refractory hyperparathyroidism and the response to vitamin D therapy in patients receiving dialysis.⁴⁶⁴ Similarly, FGF23 may be useful in the evaluation of calcium, phosphate, and vitamin D disorders in early-stage CKD in pediatric as well as adult patients.^{465,466} Lowering FGF23 levels (e.g., with oral phosphate binders) may reduce cardiovascular morbidity in CKD patients,⁴⁶⁷ although further studies are required to investigate this further. For further discussion of FGF23, see Chapter 53.

SOLUBLE UROKINASE-TYPE PLASMINOGEN ACTIVATOR RECEPTOR

Although there has long been a connection between soluble urokinase-type plasminogen activator receptor (suPAR) and CKD in the setting of focal segmental glomerulosclerosis (FSGS) and other glomerular disease, this biomarker has been associated more broadly with CKD. For more on the role of suPAR in the setting of FSGS and other glomerulonephritides, see Chapter 31. suPAR is the circulating form of a glycosyl-phosphatidylinositol-anchored, three-domain membrane protein that is expressed on a variety of cells, including endothelial cells and podocytes.⁴⁶⁸ There are circulating and bound forms of this protein, with the circulating form being produced through the cleavage of the membrane-bound form, thus making it detectable in several body fluids, including urine.^{468,469} It has been proposed that soluble urokinase-type plasminogen activator receptor (suPAR) contributes to the pathogenesis of kidney disease in the setting of FSGS as well as diabetic nephropathy through the prevention of podocyte migration and apoptosis.^{470,471} Although many of the specifics of these mechanisms remain an area of intense investigation, testing of human samples for suPAR has begun to yield interesting results.

Recently, Hayek and colleagues published a seminal paper measuring plasma suPAR levels in 3683 patients enrolled in the Emory Cardiovascular Biobank.⁴⁷² They then determined baseline and follow-up renal function in 2292 of these patients to determine the relationship between suPAR and baseline GFR, change in GFR, and development of CKD (defined as eGFR < 60 mL/min/1.73 m²). Higher baseline suPAR levels were associated with a greater decline in eGFR/year. Patients in the highest quartile of baseline suPAR lost over four times as much eGFR/year compared with those in the lowest quartile (−4.2 mL/min compared with −0.9 mL/min). Similarly, the risk of progressing to an eGFR < 60 mL/min over the study period was 3.13 (2.11–4.65) in those participants in the highest suPAR quartile compared with the lowest quartile ($n = 1335$ total patients who started with an eGFR > 60 mL/min). Although suPAR appears to be independently associated

with the development and progression of CKD, further studies are needed to validate these findings.

URINARY RENAL FIBROSIS MARKERS

Excessive production of extracellular matrix (collagen IV) and profibrotic growth factors including CTGF and TGF- β 1 have been implicated in the progression of renal fibrosis. Several reviews on the ability of renal fibrosis markers to predict patient outcomes have been published.⁴⁷³

CONNECTIVE TISSUE GROWTH FACTOR

CTGF (also known as “CCN2”), a member of CCN family of matricellular proteins, was first discovered by Bradham and colleagues in 1991 as a secreted protein in the conditioned media of human umbilical vascular endothelial cells.⁴⁷⁴ CTGF has been implicated in a variety of cellular functions, including proliferation, cell adhesion, angiogenesis, and wound healing.^{440,475,476} Accumulated evidence on CTGF in the past few years has indicated that CTGF is both a marker and a mediator of tissue fibrosis.⁴⁷⁷ CTGF is an immediate early gene potently induced by TGF- β and shown to promote fibrosis primarily through TGF- β .⁴⁷⁸ CTGF is overexpressed in several fibrotic diseases, such as scleroderma and lung and hepatic fibrosis.^{479–481} In the kidney, CTGF expression has been shown to be upregulated in various forms of renal disease, including IgA nephropathy, focal and segmental glomerulosclerosis, and diabetic nephropathy.⁴⁸⁰ CTGF has been found to be elevated in the glomeruli at early and late stages of diabetic nephropathy.⁴⁸² Riser and associates first reported that CTGF is elevated in the urine of diabetic rats and in persons with diabetes.⁴⁸³ Subsequently, several groups have reported higher urinary levels of CTGF in persons with diabetes than in healthy individuals,^{484,485} which indicates its potential as a marker for diabetic nephropathy. In persons with diabetes, plasma CTGF levels were shown to be higher in those with macroalbuminuria than in those with a normal urine albumin level. CTGF was an independent predictor of ESRD and correlated with the rate of decline in the GFR.⁴⁸⁶ In another study, both blood and urine levels of intact CTGF and the N-terminal fragment were measured in 1050 patients with type 1 diabetes from the Diabetes Control and Complications Trial/Epidemiology of Diabetes Interventions and Complication (DCCT/EDIC).⁴⁸⁷ Patients with macroalbuminuria had higher plasma levels of CTGF N-terminal fragment than those with or without microalbuminuria. Intact CTGF levels were associated with the duration of diabetes, as well as common carotid artery intima media thickness. Additionally, in regression analyses, log plasma CTGF N-terminal fragment concentrations were independently associated with the intima thickness of the common and internal carotid artery. Plasma CTGF concentration therefore serves as a risk marker for diabetic renal and vascular diseases.

More recently, urinary CTGF has been investigated for its ability to detect renal allograft fibrosis in a cohort of 315 transplant recipients.⁴⁸⁸ Following this cohort for 2 years, Metalidis and colleagues compared CTGF concentrations with protocol biopsies at 3, 12, and 24 months posttransplantation. Urinary CTGF levels were independently associated with the degree of interstitial fibrosis.⁴⁸⁸ In a subset of patients, CTGF levels at 3 months were associated with long-term moderate and severe fibrosis at the 24-month time point.

TRANSFORMING GROWTH FACTOR- β 1

TGF- β is essential for the development and differentiation of various tissues.⁴⁸⁹ Three isoforms of TGF- β have been identified in mammalian species—TGF- β 1, TGF- β 2, and TGF- β 3. TGF- β 1 is the predominant isoform in humans.⁴⁹⁰ It is mainly secreted as a high-molecular-weight inactive complex and undergoes a cleavage process for its activation.⁴⁹¹ Several studies have demonstrated the association of urine levels of TGF- β 1 with the progression of CKD. Elevated urinary TGF- β 1 levels were found in patients with glomerulonephritis and diabetic nephropathy, as well as in renal allograft recipients.^{491–494} In addition, some of the profibrotic molecules induced by TGF- β 1, including TGF- β -inducible gene H3 (*β ig-H3*) and plasminogen activator inhibitor-1 (PAI-1), were also detected at high levels in the urine.^{495,496} Because TGF- β 1 is mostly secreted as an inactive complex that requires chemical modification for its activation, *β ig-H3* and PAI-1 can be used as surrogate markers for TGF- β 1 activity. Urinary levels of both *β ig-H3* and PAI-1 have been shown to correlate with renal injury and fibrosis in patients with diabetic nephropathy.^{495,496} However, in a study of 3939 participants from the CRIC study, TGF- β levels were not shown to be significantly associated with CKD progression or the presence of macroalbuminuria.⁴⁹⁷ This is in comparison with a much smaller case-control study looking at TGF- β in the setting of pediatric obstructive nephropathy (posterior urethral valves), in which TGF- β levels were shown to be inversely correlated with the GFR.⁴⁹⁸

In the Action in Diabetes and Vascular Disease: Preterax and Diamicon Modified Release Controlled Evaluation (ADVANCE) collaborative group, the addition of serum TGF- β to traditional predictors of CKD progression (e.g., diabetes, uACR, baseline eGFR) improved the prediction of doubling of serum creatinine over a 5-year follow-up period.⁴⁹⁹ In this nested case-control study, addition of serum TGF- β significantly increased the AUC from 0.75 to 0.82 ($P < .0001$). Other more recent studies have demonstrated that urinary TGF- β can assist in determining which kidneys will recover function following unilateral obstruction.⁵⁰⁰ In a small study of 45 subjects (11 of whom did not regain function), TGF- β levels at the time of enrollment were lower in those who went on to recover function.

COLLAGEN IV

Collagen IV is a component of the extracellular matrix, and excess deposition of collagen IV is present in renal fibrosis. Furthermore, elevation of urinary collagen IV has been reported in patients with IgA nephropathy, as well as in those with diabetic nephropathy, and has been correlated with declining renal function.^{501,502} In a prospective observational cohort study of 231 normoalbuminuric and microalbuminuric patients with type 1 diabetes, urinary collagen IV was significantly associated with a decline in the eGFR over time in both univariate and multivariate analyses, with collagen IV levels being elevated in those with the lower GFRs.⁵⁰³

TUMOR NECROSIS FACTOR RECEPTORS

TNF is produced by immune cells and can either be membrane-bound or in a soluble circulating peptide.⁵⁰⁴ TNF cell surface receptors have two forms (TNFR1 and TNFR2), both of which can be cleaved from the membrane with a TNF- α cleaving enzyme. TNFR1 is expressed in glomeruli

and endothelial cells, and TNFR2 is variably expressed in renal cells.⁵⁰⁵ Increasingly, both receptors have been associated with adverse outcomes in the setting of kidney disease.^{262,506}

Pavkov and colleagues performed a longitudinal cohort study in Native Americans with type 2 diabetes, following them for a median duration of 9.5 years.⁵⁰⁶ Of a total of 193 participants, 62 developed ESRD and 25 died during the follow-up period. In an age- and gender-adjusted analysis, the highest quartile of TNFR1 and TNFR2 (compared with the lowest) was associated with an increased risk of ESRD, OR of 6.6 (3.3–13.3) and 8.8 (4.3–18.0), respectively. In more complex adjusted models (including both ACR and baseline GFR) the highest quartiles of TNFR1 and TNFR2 were associated with a 60% and 70% increased risk for ESRD compared with the lowest.⁵⁰⁶

More recently, Coca and colleagues measured TGFR1 and TGFR2 in subjects from the Action to Control Cardiovascular Risk in Diabetes (ACCORD) and Veterans Administration NEPHROPathy in Diabetes (VA-NEPHRON-D) cohorts.²⁶² They demonstrated that receptor levels were independently associated with increased decline in GFR in persons with diabetic kidney disease. In the ACCORD cohort, which enrolled subjects with early diabetic kidney disease, those who developed incident CKD had higher TNFR1 and TNFR2 levels at the time of study enrollment. Similarly, in adjusted analyses, the fourth quartiles of TNFR1 and TNFR2 (compared with lowest quartile) were both associated with increased risk of incident CKD (OR, 95 CI%, 3.0 [1.2–7.3] and 8.4 [3.0–23.4], respectively).²⁶² In the NEPHRON-D study, the fourth quartiles of TNFR1 and TNFR2 (compared with the lowest quartile) were again associated with the primary endpoint, CKD progression. The highest quartile of TNFR1 was associated with an adjusted odds of 3.5 (1.9–6.3) for the progression of CKD, and this effect was even larger for TNFR2 (3.8 [2.0–7.3]).²⁶² TNFR1 and TNFR2 were highly correlated with each other in these two cohorts (Pearson correlation coefficient > 0.75). Thus, in several cohorts with diabetic kidney disease, TNFR1 and TNFR2 levels were associated with the development and progression of CKD.

EPIDERMAL GROWTH FACTOR

As a well-established growth factor, EGF has been shown to alter the kidney's response to tubulointerstitial injury,⁵⁰⁷ with exogenous EGF improving outcomes in the setting of animal models of AKI.⁵⁰⁸ However, in a recent investigation of CKD progression using a renal biopsy transcriptome-driven approach, EGF was shown to be an excellent biomarker of CKD progression, defined as the composite of a 40% reduction from baseline eGFR or the development of ESRD.⁵⁰⁹ Intrarenal transcripts of EGF expression were correlated with baseline eGFR and patient outcomes in 164 patients from the European Renal cDNA Bank (ERCB). In this cohort, intrarenal EGF correlated with several other measures and outcomes, including urinary EGF, interstitial fibrosis and tubular atrophy, and eGFR loss. The authors were able to replicate these results in two separate CKD cohorts ($n = 55$ and $n = 48$; different samples from the ERCB cohort and Clinical Phenotyping Resource and Biobank Core [C-PROBE]).⁵⁰⁹ Adding urine EGF to a model containing age, gender, baseline eGFR, and albuminuria significantly improved the C-statistic for the prediction of CKD progression from 0.75 to 0.87. This seminal

report on EGF will be followed up with future investigations that will attempt to link this novel biomarker further with the development and incidence of progressive CKD.

COMBINATIONS OF MULTIPLE BIOMARKERS

In the classic biomarker paradigm, one biomarker detects one disease. However, acute and chronic kidney diseases are complex, with multiple underlying causes. A single biomarker may not be optimal to make an early diagnosis and predict the longer-term outcome of the disease process. Different biomarkers provide different sets of information. As discussed above, some biomarkers excel at detecting AKI in different clinical settings (cardiac surgery vs. ICU vs. ER vs. other), whereas others detect different aspects of AKI (early diagnosis vs. AKI severity vs. prerenal). This same phenomenon is true of CKD biomarkers, with some being more likely to detect CKD in the setting of diabetes versus obstruction versus other forms of glomerulonephritis. Thus, it is important to consider the clinical utility of a panel of biomarkers for acute and chronic kidney diseases.

It is becoming increasingly clear, based in part on the evidence discussed above, that multiple biomarkers are already a viable option in the care of patients with CKD. The serum creatinine level can be used in conjunction with cystatin C for the detection and diagnosis of CKD. Proteinuria and albuminuria can be used in combination with these two markers of glomerular function to diagnose and risk-stratify individuals further. Recall that in the setting of normal serum creatinine and serum cystatin C levels, the presence of albuminuria constitutes a diagnosis of CKD.

The use of multiple biomarkers in the setting of AKI has been an area of increased investigation, with several studies attempting to combine two or more biomarkers to improve their predictive capabilities.^{242,243,265,322,330} Although some studies have simply used the product of two biomarkers and then assessed the AUC, others have used logistic regression models to assess the AUC for two or more biomarkers. There has not been a consensus for the statistical methods for combining biomarkers, and this remains an area of continued investigation. More recent studies have acknowledged the aforementioned premise that individual biomarkers will have their own specific kinetics, and that combining biomarkers from different time points may improve their predictive capabilities.²⁴³ However, the clinical implications and feasibility of collecting biomarker samples at several distinct time points following cardiac surgery or ICU admission remain untested. As more and more biomarker data are amassed, we anticipate advances in novel methods for assessing biomarker combinations. Box 27.3 summarizes a variety of rationales and approaches to combining biomarker results in the hope of achieving improved prediction of patient outcomes.

CRITICAL PATH INITIATIVE: A NEED FOR BETTER BIOMARKERS

The Critical Path Initiative was launched in March 2004 by the FDA as a strategy for modernizing the sciences through which FDA-regulated products are developed, evaluated, manufactured, and used. In 2006, the Critical Path Initiative

Box 27.3 Strategies for Biomarker Combination

- Combine for different functions:
 - Combine a marker of filtration with one of tubular injury.
 - Combine a marker of proximal tubular injury with a marker of distal tubular injury.
- Combine for kinetics:
 - Combine biomarkers with different time courses to improve the duration of diagnosis.
- Combine for improved accuracy:
 - Use two or more biomarkers in statistical equations.
- Strategic combinations:
 - Combine a diagnostic biomarker with a prognostic biomarker.
 - Combine an extremely sensitive marker with an extremely specific marker.

outlined specific key areas of critical path focus identified by FDA experts and the public. Since that time, the FDA has provided guidelines stating that a biomarker can be considered “valid” only if (1) it is measured in an analytic test system, with well-established performance characteristics, and (2) there is an established scientific framework or body of evidence that elucidates the physiologic, pharmacologic, toxicologic, or clinical significance of the test result. Over the past decade there have been a few advancements in the development and approval of biomarkers of AKI and CKD.

In 2010, seven urinary proteins (KIM-1, albumin, total protein, β_2 -microglobulin, cystatin C, clusterin, and trefoil factor 3) were evaluated for their utility to outperform current tests, including serum creatinine concentration and BUN concentration, in the detection of drug-induced kidney injury. The FDA and EMA qualified this panel of biomarkers for use in regulatory decision making for drug safety to detect acute drug-induced kidney injury in preclinical studies and, on a case-by-case basis, in early clinical studies in combination with standard biomarkers.^{24,25,510,511} More recently, the FDA has approved total kidney volume in the setting of polycystic kidney disease as a marker of disease progression.²² Given the aforementioned work investigating several promising biomarkers of AKI and CKD, there will undoubtedly be additional measures approved in the next several years.

KIDNEY HEALTH INITIATIVE

In response to the epidemic of CKD and AKI and the limited number of randomized controlled trials, the FDA, American Society of Nephrology, and their industry partners announced the founding of the Kidney Health Initiative (KHI) in 2013. This public-private partnership was designed to foster collaborations to optimize the evaluation of drugs, devices, biologics and food products in the greater kidney community. This initiative is intended to facilitate the delivery of these products to the U.S. market in a safe and expeditious manner.⁵¹² Despite the growing evidence of their ability to predict AKI, CKD progression, and other adverse patient outcomes, no new biomarkers have been approved in the

United States for clinical use; as such, it remains to be seen how this new initiative will affect the biomarker field.

FUTURE OF BIOMARKERS

Recent advances in molecular analysis and proteomics have resulted in the identification of a wide range of potential serum and urine biomarkers for assessing renal function and injury, as well as predicting the development of kidney disease. Not only are many of these biomarkers sensitive, but some are also site-specific. A number of them have been reported to be predictive of an adverse outcome. For some, a great deal of additional work is still needed, however, to bring the biomarkers successfully to clinical practice.

Because kidney disease is complex, with multiple causes, and often presents in the setting of systemic diseases, a single biomarker may be insufficient for early diagnosis, insight into pathophysiology, and prediction of clinical course and outcome. Different biomarkers will be useful in different contexts. In some circumstances, a single biomarker may suffice, but in others, benefit will come from the use of multiple biomarkers in plasma, urine, or both to provide early evidence of risk and injury, and to distinguish between various types of kidney diseases. Many of these biomarkers can be grouped according to their association with a particular type of injury (e.g., podocyte or tubular injury) or mechanism of damage (e.g., oxidative stress, inflammation, fibrosis). Understanding the relationships between these different biomarker categories may help us better understand disease processes.

These biomarkers are not only useful for accessing kidney injury in humans in its early stages and for predicting progression of disease, but also crucial for translating novel therapeutic compounds from preclinical animal models to first human trials. Until recently, the use of newly emerged biomarkers in preclinical and clinical studies and drug development has been hindered by lack of regulatory acceptance. Hopefully, in the future, biomarker measurements obtained using biomarker test panels will not only be used to diagnose kidney injury and predict outcome, but will also be used as surrogate endpoints in clinical trials, which might speed up clinical evaluation of desperately needed therapies for kidney diseases.

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BOARD REVIEW QUESTIONS

- The ideal biomarker for AKI should
 - Differentiate between intrarenal, prerenal, and postrenal causes of acute kidney injury (AKI)
 - Detect AKI early in the course and be able to predict the course of disease progression
 - Be site-specific and able to correlate with histologic findings
 - Be easily and reliably measured in a minimally invasive manner
 - All of the above

Answer: e

Rationale: All of the above—the ideal marker for AKI should do all of the above.

- Creatinine is an imperfect biomarker for AKI because of all the following, except
 - Its production varies greatly with age
 - Its production varies with gender
 - It is specific for renal tubular injury
 - Static measurements of serum creatinine do not reflect real-time changes in glomerular filtration
 - Its assay is subject to interference from hyperbilirubinemia and elevated creatine kinase (CK) levels

Answer: c

Rationale: Creatinine is not specific for renal tubular injury. For example, intravascular volume depletion and prerenal factors (severe dehydration, blood volume loss, altered vasomotor tone, or age-related decrease in renal blood flow) and postrenal factors (obstruction or extravasation of urine into the peritoneal cavity) may falsely elevate serum concentrations in the absence of parenchymal damage. Thus, a decrease in the estimated glomerular filtration rate (eGFR) inferred from an increase in the serum creatinine level may not distinguish among prerenal, intrinsic renal, and postrenal causes of impaired kidney function, which may not be the case for some biomarkers of renal tubular injury.

- As a biomarker of chronic kidney disease (CKD), all the following are true about serum cystatin C, except
 - Cystatin C has been shown to be affected by gender
 - Cystatin C is lower in older adults
 - Cystatin C–based eGFRs improve CKD classification and risk stratification
 - Cystatin C values are associated with mortality

Answer: b

Rationale: Unlike serum creatinine, which is usually lower in older adults, given their decreased muscle mass, a study investigating a subset of over 7500 subjects from the NHANES III study demonstrated that cystatin C levels were elevated in more than 50% of those older than 80 years.^{89–91} Cystatin C levels have been shown to be associated with factors similar to those that are associated with creatinine—namely, these levels may be elevated in males, taller and heavier patients, and those with a higher lean body mass.^{89–91} In a prospective cohort study of 26,643 Americans enrolled in the Reasons for Geographic and Racial Differences in Stroke (REGARDS study), Peralta and colleagues have demonstrated that cystatin C–based eGFRs improve chronic kidney disease (CKD) classification and definition, as well risk stratification (for

the development of end-stage renal disease and death, relative to creatinine-based eGFRs).⁹²

- Neutrophil gelatinase associated lipocalin (NGAL) is a
 - 13-kDa, low-molecular-weight cysteine protease inhibitor produced at a constant rate by all nucleated cells and eliminated exclusively by glomerular filtration
 - 18-kDa proinflammatory cytokine that is activated by caspase 1 and is produced by renal tubule cells and macrophages
 - Type I transmembrane glycoprotein with an ectodomain containing a six-cysteine immunoglobulin-like domain, two *N*-glycosylation sites, and a mucin domain
 - 25-kDa protein glycoprotein bound to matrix metalloproteinase-9 in renal epithelial cells
 - Protein that selectively binds to free fatty acids and transports them to mitochondria or peroxisomes, where free fatty acids are beta-oxidized and participate in intracellular fatty acid homeostasis

Answer: d

Rationale: NGAL is a 25-kDa protein glycoprotein bound to matrix metalloproteinase-9 in renal epithelial cells. Cystatin C is a low-molecular-weight protein produced at a constant rate by all nucleated cells and eliminated exclusively by glomerular filtration. Interleukin 18 (IL-18) is an 18-kDa proinflammatory cytokine that is activated by caspase 1 and produced by renal tubule cells and macrophages. Kidney injury molecule-1 (KIM-1) is a type I transmembrane glycoprotein with an ectodomain containing a six-cysteine immunoglobulin-like domain, two *N*-glycosylation sites, and a mucin domain. Liver fatty acid binding protein (L-FABP) binds selectively to free fatty acids and transports them to mitochondria or peroxisomes, where free fatty acids are beta-oxidized and participate in intracellular fatty acid homeostasis.

- A 74-year-old black woman with a history of CKD (baseline creatinine, 1.4 mg/dL; eGFR, 43 mL/min/1.73 m²) and congestive heart failure (CHF), with a reduced left ventricular ejection fraction (20%), has undergone a combined a single-vessel coronary artery bypass graft (CABG) and mitral valve replacement. She had several minutes of documented hypotension in the operating room, which required transient use of norepinephrine, and had her aorta cross-clamped for 80 minutes, with a total cardiopulmonary bypass time of 120 minutes. On postoperative day 1, her creatinine was 1.9 mg/dL. The primary team decides to check the tissue inhibitor metalloproteinase-2 (TIMP-2) × insulin-like growth factor binding protein-7 (IGFBP-7) level, which returns a value of 1.2, and call a nephrology consult. Which of the following is true about the performance of this test in this patient?
 - There is no increased risk for the development of AKI
 - There is no increased risk for a composite endpoint of death or the need for dialysis in the next 9 months
 - Her baseline CKD has no impact on the test results
 - Her significant preoperative CHF (ejection fraction 20%) makes the test results difficult to interpret
 - All of the above are true

Answer: c

Rationale: Her baseline CKD has no impact on the test results. Based on data from the SAPPHERE and OPAL trials, TIMP-2 $\times$ IGFBP-7 can help risk-stratify patients for severe AKI. Patients with a value less than 0.3 were at lowest risk for severe AKI; those with a value between 0.3 and 2.0 had a relative risk 4.7 (1.5–16.0) times higher. In the SAPPHERE study, TIMP-2 $\times$ IGFBP-7 levels measured at the time of study

enrollment were also shown to correlate with a composite endpoint of death or need for renal replacement therapy over the next 9 months. In subgroup analyses, TIMP-2 $\times$ IGFBP-7 has also been shown to forecast impending severe AKI reliably in several subsets of critically ill patients, including those with advanced CKD, CHF, as well as those undergoing emergent and cardiothoracic surgery. As such, answer choice c is correct.